A power saving for planar unit distances

OpenAI

September 23, 2026

Abstract

We prove a power saving for the planar unit-distance problem: for some absolute $\beta < 4/3$, every set of n points in the Euclidean plane determines $O(n^\beta)$ unordered pairs at unit distance. This improves the exponent $4/3$ in the bound of Spencer, Szemerédi and Trotter. The proof argues by contradiction from a sequence of near-extremal configurations. Random cuttings, degree trimming and an induced bipartite subgraph maximizing its retained edge fraction divided by the 27/50-th power of the product of its two retained vertex fractions give an unbalanced point–circle incidence graph with controlled degrees and an expansion property. An entropy argument resamples a point from its answers to a short sequence of coordinate tests. After algebraic specialization, heights and the product formula over a number field containing the coordinates measure the variation of unit-edge directions over all absolute values. Comparing pairs of edges at points and centers produces a bipartite graph on two copies of a smaller point set, with a number of edges nearly quadratic in the size of the set. Each edge joins distinct neighbors of a common unit-circle center and carries the ratio of the directions from that center to its two points. These ratios have heights tending to infinity uniformly over the edges, while their alternating products around 4-cycles have uniformly negligible height in comparison. After a further deletion of pairs on unit circles containing at least $n^{3/10}$ points of the smaller set, an algebraic argument with derivations and independent square roots, using quadratic Kummer theory, rules out the resulting sequence of graphs.

1 Introduction

For a finite set $X \subset \mathbb{R}^2$, write

$$u(X) = \#\{\{x, y\} \subset X : \ \|x - y\|_2 = 1\}, \qquad u(n) = \max_{\substack{X \subset \mathbb{R}^2 \\ |X| = n}} u(X).$$

Thus pairs are unordered and their endpoints are distinct. We prove a fixed power saving over the exponent $4/3$.

Theorem 1.1. *There are absolute constants $0 < C < \infty$ and $1 \le \beta < 4/3$ such that $u(n) \le Cn^\beta$ for every nonnegative integer n.*

Theorem 1.1 gives a positive answer to the question whether this upper exponent can be lowered by a fixed amount: equivalently, $u(n) = O(n^{4/3-\delta})$ for an absolute $\delta > 0$.

1.1 The unit-distance problem and earlier methods

Erdős introduced the repeated-distance problem in 1946 [10]. Rescaling a configuration turns any specified positive distance into a unit distance, so the problem concerns the largest possible

multiplicity of one distance. His lattice construction gives the lower bound $n^{1+c/\log\log n}$ for a positive constant c and sufficiently large n [10, Theorem 2]. Spencer, Szemerédi and Trotter established the upper bound $u(n) = O(n^{4/3})$ in 1984 [25]. Székely gave a crossing-number proof and a general incidence theorem for families of curves [26, Theorems 4 and 8]. The crossing inequality was developed independently by Ajtai, Chvátal, Newborn and Szemerédi and by Leighton [3, 18]. Székely's incidence theorem applies to points and unit circles using two elementary properties: two unit circles meet in at most two points, and two points belong to at most two unit circles. Later work improved the leading constant [1] and described the structure of configurations near the extremal incidence scale [15]. Pach, Raz and Solymosi connected the problem to rigidity and obtained a logarithmic improvement conditional on a rigidity conjecture [22, Conjecture 7 and Theorem 8].

The historical unit-distance conjecture of Erdős proposed the stronger bound $u(n) = n^{1+o(1)}$. An OpenAI construction disproved it by giving at least $n^{1+\varepsilon}$ unit distances, for some fixed $\varepsilon > 0$, along an unbounded sequence of sizes [21, Theorem 1.1]. Alon et al. gave a simplified account and explained the construction's number-theoretic ancestry [4, Sections 1.2–1.3]. They credit the class-group ideas to Michel, Soundararajan, Ellenberg and Venkatesh. The construction combines these ideas with Golod–Shafarevich number-field towers having prescribed splitting, including the tower-cutting method of Hajir, Maire and Ramakrishna [9, 12, 13]. Sawin sharpened the construction to $u(n) \gg n^{1.014114}$ along an unbounded sequence of sizes [23, Theorem 1]. These are construction lower bounds. Theorem 1.1 bounds every configuration from above, leaving a gap between the known exponents.

Our proof begins with the same point–circle incidence interpretation. Random cuttings isolate a piece of the incidence graph with controlled degrees; their sampled-cell construction belongs to the framework of Clarkson and Clarkson–Shor [5, 6]. The next ingredient is information: two points sampled as neighbors of one center share little information. Entropy methods already played a role in the distinct-distance estimates of Tardos and Katz–Tardos [27, 16]. In particular, the conditional-copy argument of Katz–Tardos retains the marginal law when resampling within a class. Here that principle enters a short-history prediction lemma, proved in Section 3, in which the tests are drawn independently of the vertices and only the criterion for accepting the outcome of a test at a point may depend on a neighboring center.

The arithmetic input is the normalized product formula [19]: for a nonzero algebraic number, the sum of its logarithmic absolute values over all places of a number field containing it, with the normalized local-degree weights, is zero. We use the associated absolute height of algebraic numbers: the weighted sum of the positive parts of their logarithmic absolute values. For a unit edge, let d be the coordinate $z = x + iy$ of its point minus that of its center; the corresponding difference in the coordinate $w = x - iy$ is then d^{-1}. This reciprocity is the unit-distance equation itself, and the proof uses it, not only the two pair-counting properties of unit circles on which Székely's incidence theorem rests: points and lines satisfy those properties, with one in place of two, and attain the exponent $4/3$, so no argument that uses only them can lower it [22, Section 1.1]. For two distinct unit edges with a common endpoint and coordinate differences d, d', we have $d \neq d'$. The product formula applied to d/d' makes the sum over all places, with the same weights, of $|\log|d/d'|_v|$ twice the height of d/d' (Lemma 4.2); applied to $(d - d')(d^{-1} - d'^{-1})$, the product of the differences of the other two endpoints in the two coordinates, it gives the upper bounds for the integrated scores of pairs of edges (Proposition 5.4 and Lemma 7.2). The proof combines these bounds with prediction and, at its last stage, with algebraic independence and quadratic extensions.

1.2 The proof in outline

Suppose that no fixed power saving exists. Set $t = n^{1/3}$ along a sequence of counterexamples, so their ordered unit pairs number $t^{4-o(1)}$.

Incidence and prediction. Random cuttings and degree trimming in Section 2 give an unbalanced point–circle incidence graph with controlled degrees. Maximizing over induced bipartite subgraphs then gives expansion: small vertex sets capture few edges, and near-agreement of labels across edges forces concentration on one label on both sides. For the uniform edge law, sample a center from its marginal and then two neighbors independently; this gives an edge pair at a center. The small point side and the bound of two common centers per distinct point pair make the two point endpoints' mutual information $o(\log t)$.

Section 3 uses this information bound to resample a point conditional on its answers to a short history of coordinate tests. A history can be fixed so that the resampled point preserves acceptance by a neighboring center at an independent fresh test, up to a vanishing mean error. Resampling preserves the point marginal, while the $t^{o(1)}$ history classes make recovery of the original point unlikely. This supplies a second point for the geometric comparisons.

Height and the bounded alternative. Algebraic specialization preserves the unit equations and distinctness (Lemma 2.2), making number-field absolute values available. In coordinates $z = x + iy$, $w = x - iy$, a unit edge has coordinate differences d, d^{-1}. For an absolute value v and threshold s, record the threshold bit $\mathbf{1}_{\{-\log|d|_v > s\}}$. Let $f(v, s)$ be the minority-bit probability and $S = \int f$, integrating over thresholds with the normalized weights over places. The integrated disagreement of two edge bits is twice the height of their coordinate difference ratio.

If S remained bounded, expansion would transfer height control from edge coordinate differences to normalized point coordinates. At a fresh complex embedding, this gives positive probability of bounded, separated samples. Prediction with quantized coordinates then produces a second, separated point approximately satisfying the unit equations at three neighbors of the original point. The determinant for three separated coordinate differences forces the second point close to the first, a contradiction. Thus $S \to \infty$ (Proposition 4.3).

Comparing point and center pairs. Section 4 gives each vertex two labels recording closeness in its two coordinates; their pair is its state. Across an edge, exactly one label agrees, and the bit specifies which one, except for integrated error $O(1) = o(S)$. Separate levels with $f \geq \epsilon$ from those with $f < \epsilon$, where $\epsilon = k^{-1/16}$ for a sample size $k \to \infty$. After discarding levels where $f < \epsilon$ and the consistency error is large, expansion supplies a dominant coordinate label at each remaining level with $f < \epsilon$. A vertex's exception mass is the total weighted length of the levels where it lacks that label.

Sections 5 and 6 compare pairs of edges at a point and at a center. Their scores reward certain coordinate matches and charge bit disagreement. The product formula gives upper bounds for the integrated scores. On levels with $f \geq \epsilon$, prediction and the geometry of states agreeing in exactly one coordinate give lower bounds in terms of bit variation. A slight adjustment of the rewards near bit balance makes the lower bounds a larger multiple of the bit variation than the upper bounds. The main terms from levels with $f < \epsilon$ cancel when the two orientations are added. This forces bit disagreement on levels with $f \geq \epsilon$ in edge pairs at a center to be $o(S)$. A second comparison, using two points resampled from the same history, shows that point exceptions on

levels with $f < \epsilon$ retain a positive fraction of S; otherwise expansion would force $S = o(S)$. These are Propositions 6.1 and 6.2.

From levels with $f < \epsilon$ to ratios. On the remaining levels with $f < \epsilon$, an edge's bit is determined by its exceptional endpoints, apart from negligible errors. Subtracting the two bits in an edge pair at a center cancels the center contribution. Thus for point endpoints p, r and center q, the logarithmic absolute values of $\lambda_{pr} = (z_p - z_q)/(z_r - z_q)$ are approximately differences of functions of the absolute-value entry obtained by integrating each point's exception indicator over the retained thresholds, with sign $+1$ for minority bit 1 and -1 for minority bit 0. These functions cancel in the alternating product

$$\frac{\lambda_{pr}\lambda_{p'r'}}{\lambda_{pr'}\lambda_{p'r}}$$

around a 4-cycle.

Section 7 uses a pointwise product-formula bound and expansion to control the tail of the exception mass. It removes errors in measure weighted by exception mass before selecting a common height band. The result is a graph with at least $t^{2-o(1)}$ edges on two copies of a point set of size at most $t^{1+o(1)}$, whose ratio heights are at least $J/3$, with $J \to \infty$, and whose alternating products around 4-cycles all have height $o(J)$. Pairs on real unit circles containing at least $t^{9/10}$ points of the set can also be removed. This exclusion is needed in the last step.

Differentiation and independent square roots. Section 8 turns the separation of heights into an exact obstruction. In an ultraproduct of the number fields, elements of height $o(J)$ form a relatively algebraically closed subfield. The alternating products around 4-cycles lie in this subfield, while every edge ratio is transcendental over it. Restricting to plane or curve loci of minimal total dimension lets us choose a fixed complete bipartite pattern with positions whose coordinates have no algebraic relations over the field of locus coefficients beyond the locus equations.

A derivation vanishing on the negligible-height subfield can be chosen nonzero on every ratio in this pattern. The relations for alternating products around 4-cycles split each ratio's derivative divided by that ratio into a difference of terms attached to its endpoints. Differentiating $(z_p - z_r)(w_p - w_r) = 2 - \lambda_{pr} - \lambda_{pr}^{-1}$ therefore gives a linear system involving square roots. After one side's positions and derivative data are adjoined to the field of locus coefficients, a dimension count leaves a point on the other side with coordinate transcendence degree equal to the dimension of its locus over that enlarged field. In the function field of that locus, a separate simple zero for each radicand proves independence of their square classes. When both loci are the same unit circle the radicands are squares; this is why that case was excluded earlier.

Quadratic Kummer theory permits independent sign changes of these square roots. Solving the first few equations expresses the needed derivative terms using only those roots. Changing the sign of an unused root then forces an edge ratio's derivative divided by that ratio to vanish, contradicting its choice. This excludes every sequence approaching the exponent $4/3$, establishing a fixed positive gap.

Throughout the proof, asymptotic notation refers to the chosen sequence $t \to \infty$. Constants denoted by c, C may change between occurrences. All logarithms are natural. Every appeal to a slowly varying parameter will be made after the quantities it must dominate have been specified.

2 An incidence subgraph and its probability laws

The input to this section is a sequence of planar configurations with $t = n^{1/3} \to \infty$ and at least $t^{4-o(1)}$ ordered unit pairs. We construct an unbalanced incidence graph whose degrees are controlled on both sides. Its edge law has a uniform expansion estimate, including for arbitrarily small sets. We then prove two consequences of that estimate and show that the endpoints of an edge pair sampled at a center have mutual information $o(\log t)$.

Such an input sequence exists if Theorem 1.1 fails. Indeed, suppose that for each $j \geq 2$ some configuration of $n \geq 1$ points has more than $j^2 n^{4/3-1/j}$ ordered unit pairs. A configuration of $n < j$ points has fewer than $n^2 < j^2 n^{4/3-1/j}$ ordered unit pairs, so this configuration has $n \geq j$ points and more than $j^2 n^{4/3-1/j} \geq n^{4/3-1/j} = t^{4-3/j}$ ordered unit pairs. Passing to an increasing sequence of sizes gives the claimed input. Otherwise, for some $j \geq 2$, every configuration of $n \geq 1$ points has at most $j^2 n^{4/3-1/j}$ ordered unit pairs. Then $u(n) \leq j^2 n^{\beta}$ for every $n \geq 0$, with $\beta = \max(1, 4/3 - 1/j) < 4/3$, because $u(0) = 0$ and $n^{4/3-1/j} \leq n^{\beta}$ for $n \geq 1$; that is, Theorem 1.1 holds. We use separate copies of the configuration as points and as centers of unit circles, so its ordered unit pairs are exactly the incidences between these two copies.

Proposition 2.1. *Every sequence of configurations of $n = t^3$ distinct real planar points, with $t \to \infty$ and at least $t^{4-o(1)}$ ordered unit pairs, has replacements by distinct points with real algebraic coordinates that preserve its unit pairs, and subsets P, Q of two copies of each replacement whose unit-incidence graph contains a nonempty subgraph $E \subseteq P \times Q$ satisfying the following estimates. There is a parameter $a = a(t) > 0$ such that $a \to 0$, $a \log t \to \infty$, and*

$$|P| = t^{1+2a+o(a)}, \qquad |Q| = t^{2+a+o(a)}, \qquad |E| = t^{2+2a+o(a)}. \tag{2.1}$$

Writing $d(p), d(q)$ for graph degrees, uniformly over all vertices,

$$d(p) = t^{1+o(a)}, \qquad d(q) = t^{a+o(a)}. \tag{2.2}$$

The uniform law on E has marginals $\mu(p) = d(p)/|E|$ and $\kappa(q) = d(q)/|E|$ satisfying

$$\mu(p) \geq \frac{\gamma}{|P|}, \qquad \kappa(q) \geq \frac{\gamma}{|Q|}, \qquad \gamma = \frac{27}{50} = 0.54. \tag{2.3}$$

Set $\eta = \gamma - \frac{1}{2} = 1/25$. There is a single sequence $\xi_t \to 0$ such that, for every $F \subseteq P$ and $G \subseteq Q$, with counting fractions $u = |F|/|P|$ and $v = |G|/|Q|$,

$$\mathbb{P}_E(p \in F, q \in G) \leq \sqrt{uv}\,((uv)^{\eta} + \xi_t). \tag{2.4}$$

The same ξ_t works for all subsets F, G.

The two limits on a have different roles. The condition $a \to 0$ makes $|P| = t^{1+o(1)}$, giving the point-set size in Proposition 7.1. The condition $a \log t \to \infty$ makes the minimum center degree $t^{a+o(a)}$ tend to infinity. If a center is sampled from κ, two independent uniform neighbors coincide with probability $|Q|/|E| = t^{-a+o(a)} = o(1)$. Together with the bound of two common centers for distinct points, the point-set size and this diagonal probability give the $o(\log t)$ mutual-information estimate in Lemma 2.7. The expansion error ξ_t, which can be taken to be $t^{-a/48+o(a)}$, also tends to zero since $a \log t \to \infty$.

If Theorem 1.1 failed, the incidence subgraphs of Proposition 2.1 would be carried to a contradiction by Propositions 4.3, 6.1, 6.2, 7.1 and 8.1. The proof of Theorem 1.1 is assembled after the statement of Proposition 8.1, once all of these results have been stated.

The incidence construction is proved in Section 2.3, using algebraic specialization and the cutting below. Its matching-label and information consequences then provide the inputs to the coordinate tests.

2.1 Preserving incidences under algebraic specialization

Later arguments use number-field absolute values. The following specialization permits this without losing any of the unit pairs.

Lemma 2.2. *Let $x_1, \dots, x_n$ be distinct points of $\mathbb{R}^2$, and let $\mathcal{E}$ be any collection of pairs (i, j) for which $\|x_i - x_j\|_2 = 1$. There are distinct points $x'_1, \dots, x'_n$ with real algebraic coordinates, arbitrarily close to the respective original points, such that $\|x'_i - x'_j\|_2 = 1$ for every $(i, j) \in \mathcal{E}$.*

Proof. Fix a neighborhood of each original point and choose a rational open rectangle containing that point inside its neighborhood. For variables $X_i = (X_{i,1}, X_{i,2})$, impose membership in these rectangles, the equations

$$(X_{i,1} - X_{j,1})^2 + (X_{i,2} - X_{j,2})^2 = 1 \qquad ((i, j) \in \mathcal{E}),$$

and the strict inequalities

$$(X_{i,1} - X_{j,1})^2 + (X_{i,2} - X_{j,2})^2 > 0 \qquad (i \neq j).$$

This finite system has rational coefficients and a real solution, namely the original configuration. The transfer principle for real closed fields [17, Theorems 3.1 and 4.1] gives a solution in the field of real algebraic numbers. The equations preserve the required unit pairs, the inequalities preserve distinctness, and the rectangles give the prescribed closeness. □

We henceforth apply this lemma separately to each configuration. Extra unit pairs created by the specialization are harmless; all subsequent graphs may retain only the incidences under consideration.

2.2 Cuttings and the incidence bound

For finite sets $\mathcal{P}$ of points and $\mathcal{Q}$ of distinct unit-circle centers, let $I(\mathcal{P}, \mathcal{Q})$ denote the number of incidences. Two distinct unit circles meet in at most two points. Also, two distinct points lie on at most two unit circles: their possible centers are intersections of the two unit circles centered at the points. These two facts provide the pair-counting bounds used below. The cutting proof uses the random-sampling principle for regions with bounded defining data developed by Clarkson [5, Section 4]. We give the circle boundary conventions explicitly so that every retained point belongs to its assigned open region.

Lemma 2.3. *Let $\mathcal{P}$ be a finite planar point set and let $\mathcal{Q}$ be a nonempty set of l distinct unit circles. For $R \geq 1$, let the sample consist of $s = \lceil CR \log(2 + l) \rceil$ independent uniform draws from $\mathcal{Q}$ with replacement, where C is a sufficiently large absolute constant. With probability at least $9/10$, the points of $\mathcal{P}$ lying off the sampled circles admit an assignment to at most $C'(1 + s^2)$ open regions, each meeting at most l/R original circles, with every assigned point lying in its region. Here C' is an absolute constant, independent of the configuration, l and R.*

Proof. Choose an auxiliary horizontal direction before taking the sample. Require that an input point and an intersection point of two original circles have different horizontal coordinates unless they are the same point. Require also that an input point have a horizontal coordinate different from that of a horizontal extremum of an original circle, unless it is that extremum. Only finitely many directions are excluded. For the latter assertion, if c is a circle center and p is a point, an equality with an extremal abscissa has the form $e \cdot (p - c) = \pm 1$ for a unit horizontal direction

e, and has at most finitely many solutions in e. This auxiliary direction does not change the coordinates used elsewhere in the proof.

Ignore repetitions among the sampled circles. Express each sampled circle as its upper and lower semicircle graphs on their open horizontal domains. The event points are their horizontal extrema and all their pairwise intersections, including tangencies. Between consecutive event abscissas, the graphs have fixed vertical order; consecutive graphs, with infinite outer boundaries allowed, bound open gaps.

Sweep these gaps from left to right. Continue a gap across an event abscissa precisely when its two boundary graphs and their adjacency continue unchanged and its closed limiting vertical segment contains no event point. Otherwise start a new piece on the right. Each resulting piece is the open region between two fixed semicircle graphs, or infinite boundaries, over an open interval. It contains no part of a sampled circle. In particular, a continued gap has no start, end, tangency or crossing inside it at the abscissa across which it continues. An input point at an event abscissa must, by the choice of direction, itself be an event point and therefore lies on a sampled circle. Thus every point not deleted is in one of the open pieces.

There are $O(1+s^2)$ pieces, including in the presence of multiple intersections or several events with the same abscissa. A new right-hand gap can start only if an event point lies in its closed limiting vertical segment. If c_x distinct sampled circles pass through an event point x, only $O(1+c_x)$ right-hand gaps can have this property, by their vertical ordering. Summing over event points costs $O(s+s^2)$: extrema contribute $O(s)$, and at an intersection with $c_x \geq 2$ one may charge c_x to $2\binom{c_x}{2}$, while each pair of circles has at most two intersections. This also covers extrema that coincide with intersections. There is only a constant number of initial unbounded gaps.

Finally, all possible pieces have a description from the full circle set using a bounded number of circles. Each of their two horizontal endpoints is an extremal or intersection abscissa, or infinity; each boundary graph is specified by a circle and a sign, or is infinite. Consequently at most $C_0(1+l)^6$ candidate regions can occur, with an absolute constant C_0: the two boundary graphs have $O((1+l)^2)$ choices, and each endpoint abscissa has $O(1+l^2)$ choices. These candidates are specified before sampling. For any fixed candidate region intersecting more than l/R circles, the probability that no sampled circle intersects it is at most

$$(1-1/R)^s \leq e^{-s/R},$$

with the evident interpretation when $R=1$. An actual open piece avoids every sampled circle. A union bound over the candidate regions, with C sufficiently large, therefore gives the asserted conflict bound with failure probability at most $1/10$. □

Proposition 2.4. *For every finite set of m distinct planar points and every set of l distinct unit circles, the incidence count satisfies*

$$I \leq C\log^2(2+m+l)\big(m^{2/3}l^{2/3}+m+l\big), \tag{2.5}$$

where C is an absolute constant. In particular, when m, l are bounded by a fixed power of t, the logarithmic factor is $(\log t)^{O(1)}$.

Proof. Székely's point–curve incidence theorem [26, Theorem 8] gives the stronger bound $I = O(m^{2/3}l^{2/3}+m+l)$ when distinct curves meet in at most two points and distinct points belong to at most two curves. Distinct unit circles satisfy both hypotheses. If simple curves are taken to be arcs, cut each circle at two points outside the finite input point set; the resulting $2l$ arcs satisfy the same pair bounds and preserve every incidence. The logarithmically weakened form (2.5) follows, including the empty-set cases. □

2.3 Degree control and a maximizing induced bipartite subgraph

We now use the cutting to obtain the prescribed imbalance, and then choose an induced bipartite subgraph of the incidence graph that cannot gain too much edge density by further restriction. This last maximization supplies both expansion and lower degree bounds.

Proof of Proposition 2.1. First use Lemma 2.2. Choose $a \to 0$ sufficiently slowly that the initial incidence count is at least $t^{4-o(a)}$ and every fixed power of $\log t$ is $t^{o(a)}$. In particular $a \log t \to \infty$. Apply Lemma 2.3 with $R = t^{1-a}$ and $s = O(t^{1-a} \log t)$. There are t^3 points and circles. Incidences on the sampled circles number at most st^3, and incidences at deleted points with other circles number at most $2st^3$. For the second count, charge such an incidence to one sampled circle through its point and its incident unsampled circle. Each pair of circles receives at most two charges. Thus the deletion costs $O(t^{4-a} \log t) = o(t^{4-o(a)})$.

Each remaining region meets at most t^{2+a} circles. Further split its assigned point set into sets of size at most $2t^{1+2a}$. The total number of resulting pieces is at most the original region count plus $O(t^3/t^{1+2a})$, hence at most $t^{2-2a+o(a)}$. Some piece has at least $t^{2+2a-o(a)}$ incidences. Denote its two vertex sets temporarily by P_0, Q_0, and its edge count by e_0. At this stage

$$|P_0| \le 2t^{1+2a}, \qquad |Q_0| \le t^{2+a}, \qquad t^{2+2a-o(a)} \le e_0 \le t^{2+2a+o(a)},$$

where the upper bound follows from (2.5).

Choose $b = o(a)$ so slowly that $b \log t \to \infty$ and b dominates all exponent errors in these estimates and the logarithmic factor in (2.5). Delete point vertices of degree greater than t^{1+b}, and center vertices of degree greater than t^{a+b}, with degrees measured before these deletions. The counts of such vertices are at most

$$t^{1+2a-b/2} \quad \text{and} \quad t^{2+a-b/2},$$

respectively, for all sufficiently large t. Apply (2.5) to the excessive vertices on one side and all vertices on the other. In either case the mixed term is at most $t^{2+2a-b/3+o(b)}$; the two linear terms are bounded by $t^{1+2a+o(b)} + t^{2+a+o(b)}$. Their sum is $o(e_0)$. Both deletions therefore retain $t^{2+2a+o(a)}$ edges and give the degree upper bounds $t^{1+o(a)}$, $t^{a+o(a)}$. Dividing the retained edge count by these two upper bounds shows that both retained vertex counts attain their stated upper scales within $t^{o(a)}$. Relabel this trimmed graph as (P_0, Q_0, E_0).

For subsets of these trimmed vertex sets, let u, v be their respective counting fractions and let w be their captured edge fraction. The incidence bound and the maximum-degree bounds give, with one uniform factor $T_t = t^{o(a)}$,

$$\begin{aligned} w &\le T_t\big((uv)^{2/3} + t^{-a}v + t^{-1}u\big), \\ w &\le T_t \min(u, v). \end{aligned} \tag{2.6}$$

Taking the minimum with the degree bounds separately in the two linear terms gives

$$w \le T_t\big((uv)^{2/3} + \min(t^{-a}v, u) + \min(t^{-1}u, v)\big).$$

Indeed, $\min(A + B + C, d) \le A + \min(B, d) + \min(C, d)$ for nonnegative terms, and one can take $d = \min(u, v)$. Using $\min(x, y) \le \sqrt{xy}$ now yields

$$w \le T_t\sqrt{uv}\big((uv)^{1/6} + 2t^{-a/2}\big), \tag{2.7}$$

after increasing T_t if necessary. For example, $\min(t^{-a}v, u) \le t^{-a/2}\sqrt{uv}$; the other linear term uses $t^{-1/2} \le t^{-a/2}$ for large t.

Put $\tau = t^{-a/8}$. Among all induced bipartite subgraphs with counting fractions $u, v \ge \tau$ and captured edge fraction w, maximize

$$\frac{w}{(uv)^\gamma}, \qquad \gamma = \frac{27}{50}.$$

We choose $\gamma > 1/2$ so that a bound by $(uv)^\gamma$ improves on $\sqrt{uv}$ for small sets. The inequality $\gamma < 2/3$, where $2/3$ is the incidence exponent in (2.6), will ensure that the maximizing induced bipartite subgraph retains the scales (2.1). This is a maximum over a nonempty finite family, and its value is at least one because the whole graph is feasible. Write U, V for the maximizing counting fractions and W for its captured edge fraction. We claim

$$U, V, W \ge t^{-o(a)}. \tag{2.8}$$

Indeed (2.7) gives

$$1 \le T_t \left((UV)^{2/3-\gamma} + 2t^{-a/2}(UV)^{-\eta} \right).$$

By feasibility, $UV \ge t^{-a/4}$, so the second term on the right is at most $2t^{-49a/100+o(a)}$, which tends to zero. If $UV \le t^{-ca}$ along a subsequence for any fixed $c > 0$, the first term tends to zero as well, since $2/3 - \gamma = 19/150 > 0$. This is impossible. Thus $UV \ge t^{-o(a)}$, which implies the same for each of U, V, and $W \ge (UV)^\gamma$ gives the claim for W.

Let (P, Q, E) be this maximizing induced bipartite subgraph, retaining precisely its edges from the trimmed graph. The scale estimates (2.1) and the degree upper bounds survive by (2.8). For further subsets with fractions u, v inside P, Q, optimality gives

$$\mathbb{P}_E(F, G) \le (uv)^\gamma$$

whenever $Uu, Vv \ge \tau$. Otherwise $uv \le \tau / \min(U, V) = t^{-a/8+o(a)}$. Apply (2.7) with the old fractions Uu, Vv and divide by W. Uniformly in all such infeasible subsets, this gives

$$\mathbb{P}_E(F, G) \le t^{o(a)}\sqrt{uv}\big((uv)^{1/6} + 2t^{-a/2}\big) \le \xi_t\sqrt{uv},$$

where one may take $\xi_t = t^{-a/48+o(a)} \to 0$, with the exponent error depending only on the chosen induced bipartite subgraph and the previous uniform bounds. Combining the feasible and infeasible cases proves (2.4). Empty subsets satisfy it trivially.

Finally (2.8) implies that removing any single vertex of the maximizing induced bipartite subgraph remains feasible for all sufficiently large t. Optimality with a point p removed gives

$$1 - \mu(p) \le (1 - |P|^{-1})^\gamma \le 1 - \gamma|P|^{-1},$$

the last inequality following from concavity. The center comparison is identical. This proves (2.3), and multiplying by $|E|$ supplies the lower degree bounds in (2.2). □

In particular, the minimum degree on either side tends to infinity, and $\max_p \mu(p) = t^{-1+o(1)}$. The lower marginal bounds also convert (2.4) to the form needed for probability laws. For $h = \mu(F)$, $k = \kappa(G)$, we have $|F|/|P| \le h/\gamma$, $|G|/|Q| \le k/\gamma$, and hence

$$\mathbb{P}_E(F, G) \le C\sqrt{hk}\big((hk)^\eta + \xi_t\big), \tag{2.9}$$

with a fixed absolute C. The same ξ_t works for all subsets.

2.4 Matching labels and uniformly rare sets

We will repeatedly assign finite labels to the vertices, with no bound on the number of labels. Expansion says that if almost all edges match, one label occupies almost all the mass. A second application of expansion makes the error proportional to the actual mismatch, even when that mismatch is smaller than ξ_t.

Lemma 2.5. *For the sequence of edge laws in Proposition 2.1, there are absolute constants $s_0 > 0, C < \infty$ and an index after which the following holds uniformly for every pair of label maps $\ell_P : P \to \mathcal{A}$, $\ell_Q : Q \to \mathcal{A}$ into a finite set. If*

$$s = \mathbb{P}_E(\ell_P(p) \neq \ell_Q(q)) \leq s_0,$$

some label $a_0 \in \mathcal{A}$ satisfies

$$\mu\{\ell_P \neq a_0\} + \kappa\{\ell_Q \neq a_0\} \leq Cs.$$

Moreover, for independent $p, p' \sim \mu$, at every value of s,

$$\mathbb{P}(\ell_P(p) \neq \ell_P(p')) \leq Cs. \tag{2.10}$$

The analogous statement holds for independent centers of law κ.

Proof. Let u_i, v_i be the counting fractions of the two classes with label i, including zero fractions. Sum (2.4) over the pairs of classes with the same label. Cauchy–Schwarz gives $\sum_i \sqrt{u_i v_i} \leq 1$, and therefore

$$1 - s \leq \max_i (u_i v_i)^\eta + \xi_t.$$

For sufficiently small fixed s_0 and sufficiently large t, one label a_0 thus has both counting fractions arbitrarily close to one. Apply (2.4) to either complementary class against the whole opposite side. The marginal exception masses

$$h = \mu\{\ell_P \neq a_0\}, \qquad k = \kappa\{\ell_Q \neq a_0\}$$

are then as small as any prescribed fixed constant, uniformly over the label maps. This first conclusion need not be proportional to s.

To obtain proportionality, put $J = \mathbb{P}_E(\ell_P \neq a_0, \ell_Q \neq a_0)$. By (2.9) and $\sqrt{hk} \leq (h+k)/2$, we may choose the preceding constants so that $J \leq (h+k)/4$. The events on which exactly one endpoint has label a_0 are mismatches, so

$$s \geq h + k - 2J \geq \tfrac{1}{2}(h+k).$$

This proves the first assertion, including when $s = 0$, with no additive error. Two independent point labels differ with probability at most $2h$, giving (2.10) for small s. For $s > s_0$, the bound follows by increasing C so that $Cs_0 \geq 1$. The center assertion is identical. □

Define the *law of an edge pair at a center* on triples $(p, q, r) \in P \times Q \times P$ by first choosing $q \sim \kappa$ and then independently choosing p, r according to the conditional edge law given q. Explicitly,

$$\mathbb{P}(p, q, r) = \begin{cases} (|E| d(q))^{-1}, & (p, q), (r, q) \in E, \\ 0, & \text{otherwise.} \end{cases} \tag{2.11}$$

Each point marginal is μ, and each of the two point–center marginals is the uniform edge law.

The next lemma records how expansion suppresses joint visits to small sets: both endpoints of an edge, or both point endpoints of an edge pair at a center, lie in sets of uniformly small marginal mass only with probability a vanishing multiple of those masses. We will apply its bounds for edges and for edge pairs at a center to exceptional vertices at each level with $f < \epsilon$ in Lemma 5.2. The uniform error factors let us integrate these joint-exception bounds even when the exceptional sets vary with the level.

Lemma 2.6. *Let $\delta_t \to 0$ be any nonnegative sequence. The estimates below hold uniformly over all indicated subsets, with error factors depending only on δ_t, ξ_t and the fixed constants in (2.9):*

(i) *If $F \subseteq P$, $G \subseteq Q$, and $h = \mu(F) \le \delta_t$, while $k = \kappa(G)$ is arbitrary, then*

$$\mathbb{P}_E(F, G) \le C(\delta_t^{\eta} + \xi_t)\sqrt{hk} = o(1)\sqrt{hk}. \tag{2.12}$$

The same assertion holds with the sides interchanged.

(ii) *If also $k \le \delta_t$, then*

$$\mathbb{P}_E(F, G) \le \tfrac{C}{2}(\delta_t^{2\eta} + \xi_t)(h + k) = o(1)(h + k). \tag{2.13}$$

(iii) *If $F \subseteq P$ and $h = \mu(F) \le \delta_t$, then under the law of an edge pair at a center,*

$$\mathbb{P}(p \in F, r \in F) = o(1)h. \tag{2.14}$$

In particular, the error factors may be taken outside sums or integrals over any family of sets whose marginal masses obey the same uniform bound δ_t.

Proof. The first two assertions follow directly from (2.9), using $hk \le \delta_t$ in the first case and $hk \le \delta_t^2$ and $\sqrt{hk} \le (h + k)/2$ in the second.

For the third, let $g(q) = \mathbb{P}(p \in F \mid q)$, so that $\mathbb{E}_\kappa g = h$ and the probability in question is $\mathbb{E}_\kappa g^2$. There is nothing to prove when $h = 0$. For any $d \in (0, 1]$, put $G_d = \{q : g(q) > d\}$. Markov's inequality gives $\kappa(G_d) \le h/d$. Splitting at this threshold,

$$\mathbb{E}_\kappa g^2 \le dh + \mathbb{E}_\kappa[g\mathbf{1}_{G_d}] = dh + \mathbb{P}_E(F, G_d) \le h\left[d + Cd^{-1/2}\big((\delta_t^2/d)^{\eta} + \xi_t\big)\right].$$

Choose $d = d_t \to 0$ sufficiently slowly that $\delta_t^{2\eta} d_t^{-1/2-\eta} \to 0$ and $\xi_t d_t^{-1/2} \to 0$. The bracket then tends to zero, independently of F. This proves the uniform assertion and hence its integrated form. □

2.5 The information carried by an edge pair at a center

The small point side and the bound of two common centers together ensure that the two point endpoints of an edge pair at a center share little information. We record the atom estimates as well, since they later convert a probability of edge pairs at a center into a count of distinct point pairs.

For probability laws a, b on the same finite set, define their relative entropy by

$$\mathrm{D}_{\mathrm{KL}}(a\|b) = \sum_x a(x) \log \frac{a(x)}{b(x)}, \tag{2.15}$$

with a zero summand when $a(x) = 0$ and value $+\infty$ if $a(x) > 0 = b(x)$.

Lemma 2.7. *For the incidence subgraphs of Proposition 2.1 and their law of an edge pair at a center (2.11), let $d_{\min,Q} = \min_{q\in Q} d(q)$. If $p \neq r$, then*

$$\mathbb{P}(p,r) \le \frac{2}{|E| d_{\min,Q}} = t^{-2-3a+o(a)}. \tag{2.16}$$

The diagonal probability is

$$\mathbb{P}(p=r) = \frac{|Q|}{|E|} = t^{-a+o(a)} = o(1). \tag{2.17}$$

The mutual information of the point endpoints satisfies

$$D := \mathrm{D}_{\mathrm{KL}}(\mathcal{L}(p,r) \,\|\, \mu \otimes \mu) = o(\log t). \tag{2.18}$$

Proof. For fixed p, r, summing (2.11) gives

$$\mathbb{P}(p,r) = \frac{1}{|E|} \sum_{q:(p,q),(r,q)\in E} \frac{1}{d(q)}.$$

There are at most two summands when $p \neq r$, proving the off-diagonal estimate by (2.1)–(2.2). Summing instead over the diagonal yields

$$\mathbb{P}(p=r) = \frac{1}{|E|} \sum_{q\in Q} \frac{d(q)}{d(q)} = \frac{|Q|}{|E|}.$$

The asymptotics follow from the same estimates and $a \log t \to \infty$.

For a finite random variable X, write $\mathsf{H}(X) = -\sum_x \mathbb{P}(X=x) \log \mathbb{P}(X=x)$, with $0 \log 0 = 0$. Every off-diagonal atom is at most $M_t = 2/(|E| d_{\min,Q})$. Dropping the nonnegative diagonal entropy contributions therefore gives

$$\mathsf{H}(p,r) \ge (1 - \mathbb{P}(p=r)) \log(1/M_t) \ge (2 - o(1)) \log t.$$

On the other hand, $\mathsf{H}(p) = \mathsf{H}(r) \le \log |P| = (1 + o(1)) \log t$. Expanding the definition of relative entropy gives

$$0 \le D = \mathsf{H}(p) + \mathsf{H}(r) - \mathsf{H}(p,r) = o(\log t),$$

as required. □

We retain this graph and these probability laws for the rest of the proof: the uniform edge law, its point and center marginals, and the law of an edge pair at a center. The expansion consequences hold for every finite labeling and every sufficiently small exceptional set, with constants independent of any tests subsequently assigned to the vertices.

3 Prediction from short histories

The incidence reduction supplies a probability law on point–center pairs whose point endpoints in a random edge pair at a center have mutual information $D = o(\log t)$. We now turn this information bound into predictions from short histories of point tests. The prediction uses only the history of the point; its accuracy is measured at a fresh test, drawn independently of the vertices, by a criterion that may depend on the center. We first prove this statement for an arbitrary finite

probability law and then give the two forms needed for coordinate quantization and coordinate equality.

The proof sums the information contributed by successive tests. Since that sum is at most D, a sufficiently long history leaves little new dependence on average. Related short conditioning arguments occur in Ahlswede's wringing method [2, Section 4, Lemma 3]. Here the proof must additionally compare each endpoint with its own posterior law and leave the next test independent of the selected history.

3.1 The probability laws and the entropy comparison

Let P, Q be finite nonempty sets and let $\omega(p, q)$ be a probability mass function on $P \times Q$, with marginals μ and κ. Delete atoms of zero marginal mass. Write

$$\omega(p \mid q) = \frac{\omega(p, q)}{\kappa(q)}, \qquad \omega(q \mid p) = \frac{\omega(p, q)}{\mu(p)}.$$

The associated law of an edge pair at a center and its point-pair marginal are

$$\Omega(p, q, r) = \kappa(q)\omega(p \mid q)\omega(r \mid q), \qquad \pi(p, r) = \sum_{q \in Q} \Omega(p, q, r). \tag{3.1}$$

Both marginals of π are μ. Relative entropy is defined in (2.15). For finite probability laws a, b, set

$$\mathrm{TV}(a, b) = \frac{1}{2} \sum_x |a(x) - b(x)|.$$

The information available to the argument is

$$D = \mathrm{D}_{\mathrm{KL}}(\pi \| \mu \otimes \mu). \tag{3.2}$$

It is finite, since $\pi(p, r) > 0$ implies $\mu(p)\mu(r) > 0$.

We use the standard chain rule for relative entropy [8, Theorem 2.5.3]:

$$\mathrm{D}_{\mathrm{KL}}(a_{XY} \| b_{XY}) = \mathrm{D}_{\mathrm{KL}}(a_X \| b_X) + \sum_x a_X(x)\, \mathrm{D}_{\mathrm{KL}}(a_{Y|x} \| b_{Y|x}). \tag{3.3}$$

Nonnegativity and the chain rule imply that applying a deterministic map cannot increase relative entropy: adjoin the map to the outcome and then discard the nonnegative conditional term. We also use Pinsker's inequality [8, Lemma 11.6.1]. The source uses base-two logarithms and the distance $\sum_x |a(x) - b(x)|$. Multiplying its relative entropy by $\log 2$ converts to natural logarithms; the distance is $2\,\mathrm{TV}(a, b)$. Thus for arbitrary finite laws a, b,

$$2\,\mathrm{TV}(a, b)^2 \le \mathrm{D}_{\mathrm{KL}}(a \| b). \tag{3.4}$$

Finally, total variation controls bounded expectations by

$$|\mathbb{E}_a F - \mathbb{E}_b F| \le 2\|F\|_\infty\, \mathrm{TV}(a, b). \tag{3.5}$$

A test T is a random element of a probability space, independent of (p, q, r), which specifies a code map $c_T : P \to [M]$, where $[M] = \{1, \dots, M\}$. The test space may be infinite; all maps and integrands below are assumed measurable. Only finite probability laws of codes are used. All vector norms and inner products in this section are Euclidean. For a fixed history map $H : P \to \mathcal{H}$, put

$$\mu_h(x) = \frac{\mu(x)\mathbf{1}_{\{H(x)=h\}}}{\mu(H = h)} \quad \text{when } \mu(H = h) > 0. \tag{3.6}$$

Its values on histories of zero mass are immaterial.

Lemma 3.1 (Prediction device)**.** *Let ω, π, D be as in (3.1)–(3.2), let $K, M \geq 1$ be integers, and let $T_1, \ldots, T_K$ be independent tests with the same law as T. Each test specifies, in addition to c_T, a feature map $f_T : [M] \to \mathbb{R}^d$ and a coefficient map $g_T : Q \to \mathbb{R}^d$, for an integer $d \geq 1$ and constants $B, B' \geq 0$, satisfying*

$$\|f_T(c)\| \leq B, \qquad \|g_T(q)\| \leq B'$$

for all codes, centers and tests. Let J be uniform on $[K]$ and independent of the tests, and let H be the random strict-prefix map

$$H(p) = (c_{T_1}(p), \ldots, c_{T_{J-1}}(p)).$$

For a fresh test T, independent of $J, T_1, \ldots, T_{J-1}$, put

$$m_{T,h} = \sum_{x \in P} \mu_h(x) f_T(c_T(x)).$$

Then

$$\mathbb{E}_{J,T_1,\ldots,T_{J-1},T} \left| \mathbb{E}_{(p,q)\sim\omega} \langle f_T(c_T(p)) - m_{T,H(p)}, g_T(q) \rangle \right| \leq 4BB' \left(\frac{D}{K} \right)^{1/4}. \tag{3.7}$$

Every realized history has at most M^{K-1} values. In particular, a history can be fixed before drawing the fresh test so that its average absolute discrepancy in (3.7) is at most the right side.

Proof. First freeze all K tests. Apply their complete transcript map to both coordinates of the pair (p, r). The actual transcript law is the image of π; the reference law is the image of $\mu \otimes \mu$. Data processing bounds their relative entropy by D. For $j \in [K]$, write H_j for the transcript strictly before step j. Conditional on prefixes (h, h'), the reference next-code law is exactly

$$b_{j,h,h'} = (c_{T_j})_* \mu_h \ \otimes \ (c_{T_j})_* \mu_{h'}.$$

Here μ_h is formed with H_j. Let $a_{j,h,h'}$ denote the actual conditional law of the next two codes. All prefix pairs of positive actual probability have positive reference probability. Iterating (3.3) therefore gives

$$\sum_{j=1}^{K} \mathbb{E}_{(p,r)\sim\pi} \, \mathrm{D}_{\mathrm{KL}}\big(a_{j,H_j(p),H_j(r)} \| b_{j,H_j(p),H_j(r)}\big) \leq D. \tag{3.8}$$

Consequently, by (3.4) and Cauchy–Schwarz,

$$\frac{1}{K} \sum_{j=1}^{K} \mathbb{E}_{(p,r)\sim\pi} \, \mathrm{TV}\big(a_{j,H_j(p),H_j(r)}, b_{j,H_j(p),H_j(r)}\big) \leq \sqrt{\frac{D}{2K}}. \tag{3.9}$$

For this fixed test sequence and step j, subtract the prediction belonging to each point's own prefix:

$$U_j(p) = f_{T_j}(c_{T_j}(p)) - \sum_x \mu_{H_j(p)}(x) f_{T_j}(c_{T_j}(x)).$$

The norm is at most $2B$. Under each reference conditional product, both residuals are centered and independent, so the conditional expectation of $\langle U_j(p), U_j(r) \rangle$ is zero. This inner product has absolute value at most $4B^2$. Thus (3.5) and (3.9) imply

$$\frac{1}{K} \sum_{j=1}^{K} \mathbb{E}_{(p,r)\sim\pi} \langle U_j(p), U_j(r) \rangle \leq 8B^2 \sqrt{\frac{D}{2K}}. \tag{3.10}$$

Each summand is nonnegative, for conditional independence in the law of an edge pair at a center gives the exact identity

$$\mathbb{E}_{(p,r)\sim\pi}\langle U_j(p), U_j(r)\rangle = \mathbb{E}_{q\sim\kappa}\left\|\mathbb{E}_{p\sim\omega(\cdot|q)}U_j(p)\right\|^2. \tag{3.11}$$

It follows, by Cauchy–Schwarz first over q and then over the tests and step, that the average absolute discrepancy is at most

$$B'\left(8B^2\sqrt{\frac{D}{2K}}\right)^{1/2} \le 4BB'(D/K)^{1/4}.$$

Everything proved so far holds after averaging the initially frozen tests. The residual at step j uses only $T_1,\dots,T_j$. Since the tests are independent and identically distributed, its joint law with the strict prefix is unchanged when T_j is replaced by a fresh test after that prefix. This proves (3.7). Finally average over the strict prefixes and choose one attaining at most the average. There are at most $M^{j-1} \le M^{K-1}$ possible histories for a prefix of length $j-1$. □

The important comparison in this proof is conditional on two possibly different histories. The reference law uses the first endpoint's prediction at the first coordinate and the second endpoint's prediction at the second. Their independence is used only conditional on the center, in (3.11).

3.2 Acceptance tests and the prediction coupling

Corollary 3.2 (Finite-code acceptance)**.** *Let ω,π,D be as in (3.1)–(3.2), and let $K,M\ge 1$ be integers. Suppose an independent test T specifies a code map $c_T : P\to[M]$ and accepted sets $\mathcal{A}_T(q)\subseteq[M]$ for $q\in Q$. Histories are strict prefixes of K independent tests with this law. Put*

$$s_0 = \mathbb{E}_T\mathbb{P}_{(p,q)\sim\omega}\{c_T(p)\notin\mathcal{A}_T(q)\}.$$

There is a fixed strict-prefix history H, with at most M^{K-1} values, such that

$$\mathbb{E}_T\sum_{p,q}\omega(p,q)\sum_x \mu_{H(p)}(x)\mathbf{1}_{\{c_T(x)\notin\mathcal{A}_T(q)\}} \le s_0 + 4\sqrt{M}(D/K)^{1/4}. \tag{3.12}$$

The fresh test T is independent of the fixed history construction.

Proof. Use $f_T(c)=e_c\in\mathbb{R}^M$, where e_c is the c-th standard basis vector and has norm one, and use the acceptance indicator vector as $g_T(q)$, of norm at most $\sqrt{M}$. Actual and predicted failure probabilities differ by the negative of their acceptance probabilities. Averaging (3.7) and then fixing a strict prefix proves the assertion. □

We record the sampling laws used with this result. For any fixed history H and integer $r\ge 1$, draw one point, one prediction, and r neighbors according to

$$\mathbb{P}\{p,\widehat{p},q_1,\dots,q_r\} = \mu(p)\mu_{H(p)}(\widehat{p})\prod_{j=1}^{r}\omega(q_j\mid p), \tag{3.13}$$

and draw the fresh test independently of this entire tuple. The two point marginals are both μ, each (p,q_j) has law ω, and each q_j has marginal κ. In particular each pair $(p,\widehat{p})$ is independent of the

fresh test. These assertions follow by summing (3.13), using $\sum_{p:H(p)=h} \mu(p)\mu_h(x) = \mu(x)\mathbf{1}_{\{H(x)=h\}}$. For every fixed history,

$$\mathbb{P}\{p = \widehat{p}\} = \sum_h \frac{\sum_{p:H(p)=h} \mu(p)^2}{\mu(H=h)} \le |H(P)| \max_p \mu(p). \tag{3.14}$$

Given $H(p)$, the prediction is independent of all neighbors in (3.13). Applying a union bound to (3.12) therefore bounds the chance that at least one of the r acceptance comparisons rejects this *same* predicted point at the same fresh test by $r(s_0 + 4\sqrt{M}(D/K)^{1/4})$.

There is also a useful equivalent way to draw the point pair: draw h with probability $\mu(H=h)$, then draw two points independently from μ_h. This again has both marginals μ, and its collision probability satisfies (3.14). Once H is fixed, this pair law can be used unchanged for every fresh test. Sampling a second object from the first object's fiber also appears in the entropy argument of Katz and Tardos [16, Section 2, proof of Lemma 1]; the identities above specify the marginals and independent test needed here.

For the coordinate-quantization application, a test will be an independently chosen field embedding, a code will specify bounded coordinate bins together with an overflow code, and $\mathcal{A}_T(q)$ will consist of codes achieved by actual neighbors of q. The exact hypotheses needed from that construction are a fixed finite alphabet of size M and acceptance of every actual edge, so $s_0 = 0$. If $K = o(\log t)$, $D/K = o(1)$, and $\max_p \mu(p) \le t^{-1+o(1)}$, then the prediction failure and the collision probability both tend to zero. Formula (3.13) with $r = 3$ is the simultaneous three-neighbor experiment. Geometric estimates valid for all couplings with these marginals remain valid for each fixed history. More generally, additional nonnegative error quantities can be averaged together with the prediction error before choosing a prefix; this always leaves the test fresh.

3.3 Equality of one coordinate

The next application allows an arbitrarily large set of state labels without paying for its size in the feature norm. A state is a pair in a product $\mathcal{A} \times \mathcal{B}$. For a state $y = (a, b)$, define the set of states agreeing with it in exactly one coordinate

$$R_y = \{(a', b') : (a' = a \text{ and } b' \ne b) \text{ or } (a' \ne a \text{ and } b' = b)\}. \tag{3.15}$$

Thus exactly one coordinate agrees. A fresh test specifies state maps $\mathbf{y}_T(x) = (A_T(x), B_{1,T}(x))$ for all $x \in P \cup Q$. The subscript in B_1 is only part of this label's name. Only the finitely many labels attained on these vertex sets need be retained. The labels and their equality relations are assumed measurable in T.

Corollary 3.3 (Prediction of agreement in exactly one coordinate)**.** *Let ω, π, D be as above, let $K \ge 1, L \ge 2$ be integers, and let T be an independent random test giving the state maps just described. Set*

$$s_0 = \mathbb{E}_T \mathbb{P}_{(p,q)\sim\omega}\{\mathbf{y}_T(p) \notin R_{\mathbf{y}_T(q)}\}.$$

There exists a fixed history $H : P \to \mathcal{H}$, with $|H(P)| \le L^{2K}$, for which the state laws

$$\nu_{h,T} = (\mathbf{y}_T)_* \mu_h$$

satisfy

$$\mathbb{E}_T \mathbb{E}_{(p,q)\sim\omega}[1 - \nu_{H(p),T}(R_{\mathbf{y}_T(q)})] \le s_0 + \frac{4}{L} + 18(D/K)^{1/4}. \tag{3.16}$$

The history is formed from earlier tests and does not use T.

Proof. Augment each test by two fresh independent random maps from its attained coordinate labels into $[L]$, assigning independent uniform values to distinct labels. Numbering attained labels by their first occurrence in a fixed vertex order makes this construction measurable. Use the same map on both vertex sets for each coordinate, and use independent maps for the two coordinates and for different tests. The point code is its two mapped coordinates, so the alphabet has size $M = L^2$. For a code $(a, b) \in [L]^2$, use the feature

$$f(a,b) = (e_a, e_b, e_{(a,b)}) \in \mathbb{R}^L \oplus \mathbb{R}^L \oplus \mathbb{R}^{L^2}, \qquad \|f(a,b)\| = \sqrt{3}.$$

For the mapped center state (a_q, b_q), use

$$g(q) = (e_{a_q}, e_{b_q}, -2e_{(a_q,b_q)}), \qquad \|g(q)\| = \sqrt{6}.$$

Their inner product is $\mathbf{1}_{\{a=a_q\}} + \mathbf{1}_{\{b=b_q\}} - 2\mathbf{1}_{\{(a,b)=(a_q,b_q)\}}$, exactly the indicator of agreement of exactly one mapped coordinate. Therefore Lemma 3.1 compares its actual and predicted probabilities with error at most $4\sqrt{18}(D/K)^{1/4}$, averaged over strict prefixes and the fresh augmented test.

For any two fixed true states, the random maps preserve an equality, and create an equality between two unequal coordinate labels with probability $1/L$. Thus the indicator of agreement in exactly one true coordinate differs from its mapped indicator with probability at most $2/L$. This applies both to the actual state pair and to the predicted pair: conditional on the earlier prefix, the fresh true test and the chosen vertices, the current random maps are still independent uniform maps. In particular the predicted point is sampled by (3.6) using no current random map. Comparing actual true, actual mapped, predicted mapped, and predicted true failures gives, in that order, total extra error at most $4/L + 4\sqrt{18}(D/K)^{1/4}$. Average the current test and current random maps before fixing a prefix, and use $4\sqrt{18} < 18$. The resulting true-state expression no longer involves the current random maps. Its history has at most $(L^2)^{K-1} \le L^{2K}$ values. □

One form of Corollary 3.3 that will be convenient is an integrated version. Let $\mathcal{T}$ be a measurable family of state tests with finite measure W, and let $m(\tau)$ bound failure of agreement in exactly one coordinate at τ. For $W > 0$, sample T from the normalized measure $\mathrm{d}\tau/W$. The resulting fixed history satisfies

$$\int_{\mathcal{T}} \mathbb{E}_{(p,q)\sim\omega}[1 - \nu_{H(p),\tau}(R_{\mathbf{y}_\tau(q)})]\,\mathrm{d}\tau \le \int_{\mathcal{T}} m(\tau)\,\mathrm{d}\tau + W\,(4/L + 18(D/K)^{1/4}). \tag{3.17}$$

For $W = 0$, the assertion holds with the empty history. In the later level application, $\mathcal{T}$ is the set of levels with $f \ge \epsilon$, equipped with its weighted level measure, and m is the incidence inconsistency probability. Consistency implies agreement in exactly one coordinate, which is the only geometric hypothesis needed for (3.17). We suppress the test subscript and write ν_h when the level being tested is fixed.

3.4 Choosing the history length

There is no conflict between small prediction error and a subpolynomial number of histories. Write $T_0 = \log t$, and suppose $D = o(T_0)$. For example, take

$$K = \left\lceil \sqrt{(D+1)T_0} \right\rceil.$$

Then $K \to \infty$, $K = o(T_0)$, and $D/K \to 0$. For a fixed alphabet this gives $M^K = t^{o(1)}$. For the random maps one may additionally take

$$L = \left\lfloor \exp \sqrt{T_0/K} \right\rfloor .$$

Eventually $L \geq 2$, $L \to \infty$, and $K \log L = o(T_0)$, so $L^{2K} = t^{o(1)}$. If a later argument requires an integer $k \to \infty$ satisfying $kK \log L = o(T_0)$ and $k^{20}\eta_t \to 0$ for an additional positive sequence $\eta_t \to 0$, it suffices to choose

$$k = \left\lfloor \min \left\{ \left(\frac{T_0}{K \log L} \right)^{1/2}, \eta_t^{-1/40} \right\} \right\rfloor .$$

Further finitely many slow-growth requirements can be imposed by taking a smaller such diverging sequence. Thus the prediction lemma supplies both vanishing error and the small history counts required in the subsequent geometric arguments.

4 Heights and coordinate tests

We apply the prediction device to the incidence subgraphs of Proposition 2.1. Their coordinates are real algebraic numbers. The first task is to measure the variation of edge directions simultaneously at all absolute values of a number field. We will prove that this variation cannot remain bounded. We then construct coordinate labels whose agreement records a threshold comparison, with a bounded total error.

Write

$$z_x = x_1 + ix_2, \qquad w_x = x_1 - ix_2$$

for a point or center $x = (x_1, x_2)$. For every edge $e = (p, q)$, put

$$d_e = z_p - z_q, \qquad w_p - w_q = d_e^{-1}. \tag{4.1}$$

The second equality follows from the unit-distance equation, and in particular $d_e \neq 0$. Distinct real points have distinct z-coordinates and distinct w-coordinates. These nonzero differences remain nonzero under every field embedding. At an embedding other than the original one, we use the two embedded coordinates separately.

4.1 A weighted product formula

For each configuration choose a finite normal number field $\mathcal{N}$ containing all its coordinates and i, and let $d_{\mathcal{N}} = [\mathcal{N} : \mathbb{Q}]$. The following construction allows repeated absolute values; the repetitions make their weights particularly simple.

Lemma 4.1 (Weighted absolute values)**.** *The finite normal number field $\mathcal{N}$ admits a countable collection of absolute-value entries v, each of weight $1/d_{\mathcal{N}}$, with the following properties. Its archimedean entries are $|u|_v = |\sigma(u)|$ for the $d_{\mathcal{N}}$ embeddings $\sigma : \mathcal{N} \to \mathbb{C}$, so their total weight is one. All other entries are nonarchimedean. For each nonzero $u \in \mathcal{N}$, only finitely many nonarchimedean entries have $|u|_v \neq 1$, and*

$$\sum_v{}^{*} \log |u|_v = 0. \tag{4.2}$$

Here and below a star denotes the sum with the stated weights.

Proof. Use the classification of number-field places and the product formula [19, Theorems 7.14–7.15]. At a finite place $\mathfrak{p}$ above a rational prime ℓ, normalize the genuine absolute value by $|\ell|_{\mathfrak{p}} = \ell^{-1}$, and put $n_{\mathfrak{p}} = [\mathcal{N}_{\mathfrak{p}} : \mathbb{Q}_\ell]$, where $\mathcal{N}_{\mathfrak{p}}$ is the completion. Use $n_{\mathfrak{p}}$ copies of this absolute value, each of weight $1/d_{\mathcal{N}}$. Milne's local normalization formula [19, Lemma 8.6] converts his finite-place value to $|\cdot|_{\mathfrak{p}}^{n_{\mathfrak{p}}}$. His complex-place value is squared modulus, so we instead use ordinary modulus once for each embedding, counting conjugate embeddings separately. Taking logarithms of the product formula gives, for every $u \in \mathcal{N}^\times$,

$$\sum_v{}^{*} \log |u|_v = \frac{1}{d_{\mathcal{N}}} \left(\sum_{\mathfrak{p}} \log(|u|_{\mathfrak{p}}^{n_{\mathfrak{p}}}) + \sum_{\{\sigma,\overline{\sigma}\}} \log |\sigma(u)|^2 \right) = 0.$$

The second sum is over conjugate pairs; there are no real embeddings because $i \in \mathcal{N}$. The values used in the manuscript are ordinary moduli and genuine nonarchimedean absolute values, so the triangle inequalities also hold before taking these weighted sums. There are exactly $d_{\mathcal{N}}$ embeddings, giving archimedean weight one. Finally, the principal fractional ideal of any nonzero u has finitely many prime factors [19, Theorem 3.20]; at every other finite place $|u|_v = 1$. □

Define $\log^+ r = \max(0, \log r)$ for $r > 0$ and $\log^+ 0 = 0$. Set $h(0) = 0$, and put

$$h(u) = \sum_v{}^{*} \log^+ |u|_v \qquad (u \neq 0).$$

Since the entries of Lemma 4.1 list each place with multiplicity equal to its local degree, $h(u)$ is the absolute logarithmic (Weil) height of u, that is, of the point $[1 : u]$ [24, slide 3]; Lemma 4.2 records the properties of it that we use.

Lemma 4.2 (Height inequalities)**.** *The value $h(u)$ is independent of the chosen finite normal field and of the extensions of the prime absolute values in Lemma 4.1. For $u \in \mathcal{N}^\times$,*

$$h(u^{-1}) = h(u), \qquad \sum_v{}^{*} |\log |u|_v| = 2h(u). \tag{4.3}$$

For arbitrary $u, v \in \mathcal{N}$,

$$h(uv) \leq h(u) + h(v), \qquad h(u+v) \leq h(u) + h(v) + \log 2. \tag{4.4}$$

If a random $X \in \mathcal{N}$ is independent of a uniformly sampled embedding $\sigma : \mathcal{N} \to \mathbb{C}$, then, for $B > 1$ and $0 < r < 1$,

$$\mathbb{P}\{|\sigma(X)| > B\} \leq \frac{\mathbb{E}h(X)}{\log B}, \tag{4.5}$$

$$\mathbb{P}\{X \neq 0,\ |\sigma(X)| < r\} \leq \frac{\mathbb{E}h(X)}{\log(1/r)}. \tag{4.6}$$

Proof. For a finite extension $L/\mathcal{N}$, the decomposition into completions [19, Proposition 8.2] gives $\sum_{\mathfrak{q}|\mathfrak{p}} [L_{\mathfrak{q}} : \mathcal{N}_{\mathfrak{p}}] = [L : \mathcal{N}]$. Hence

$$\sum_{\mathfrak{q}|\mathfrak{p}} \frac{[L_{\mathfrak{q}} : \mathbb{Q}_\ell]}{[L : \mathbb{Q}]} = \frac{[\mathcal{N}_{\mathfrak{p}} : \mathbb{Q}_\ell]}{[\mathcal{N} : \mathbb{Q}]}.$$

The extended absolute values restrict to the chosen normalization on $\mathcal{N}$. In particular, for every $u \in \mathcal{N}$,

$$\sum_{\mathfrak{q}|\mathfrak{p}} \frac{[L_{\mathfrak{q}} : \mathbb{Q}_\ell]}{[L : \mathbb{Q}]} \log^+ |u|_{\mathfrak{q}} = \frac{[\mathcal{N}_{\mathfrak{p}} : \mathbb{Q}_\ell]}{[\mathcal{N} : \mathbb{Q}]} \log^+ |u|_{\mathfrak{p}}.$$

Thus each finite-place contribution to height is unchanged. Each embedding of $\mathcal{N}$ has $[L : \mathcal{N}]$ extensions, giving the same conclusion at infinity. A common finite normal overfield compares any two field choices. The place list includes all normalized extensions with their local-degree multiplicities, so their enumeration involves no further choice. This proves height independence; $h(0) = 0$ is fixed.

The product formula equates the sums of positive and negative logarithms, giving (4.3). Multiplication gives the first height inequality termwise. The triangle inequality contributes at most $\log 2$ at an archimedean entry and no constant at a nonarchimedean entry; the total archimedean weight is one. This proves the second inequality, including zero summands by the stated convention. For the tail estimates, the average positive logarithm over embeddings is at most $h(X)$. For $X \neq 0$, its average negative logarithm is at most $h(X^{-1}) = h(X)$. Markov's inequality proves both assertions. □

4.2 Thresholds and their integrated minority-bit probability

For every entry v define

$$n_e(v) = -\log |d_e|_v.$$

Choose a finite closed interval $I_v = [\ell_v, u_v]$ containing 0 and all $n_e(v)$, enlarged so that for distinct $p, p' \in P$,

$$\ell_v \leq -\log |z_p - z_{p'}|_v, \qquad u_v \geq \log |w_p - w_{p'}|_v. \tag{4.7}$$

All the differences here are nonzero. The enlargement (4.7) is used only in the proof of Proposition 6.2, where it allows an integral over the level interval to be compared with the logarithms of coordinate differences of two points. By Lemma 4.1, we can take $I_v = [0, 0]$ at all but finitely many entries. Define the weighted threshold space $\mathcal{S}$ to be the disjoint union of the $\{v\} \times I_v$, with weighted Lebesgue measure:

$$\int_{\mathcal{S}} g := \sum_v^* \int_{\ell_v}^{u_v} g(v, s)\, \mathrm{d}s.$$

We usually suppress the domain $\mathcal{S}$ in an integral. At a level (v, s) put

$$\begin{aligned} Y_e(v, s) &= \mathbf{1}_{\{n_e(v) > s\}}, & \varrho(v, s) &= \mathbb{P}_E\{Y_e(v, s) = 1\}, \\ f(v, s) &= \min(\varrho(v, s), 1 - \varrho(v, s)), & S &= \int f. \end{aligned} \tag{4.8}$$

Enlarging any I_v further does not change S, because all edge bits agree outside the range of the $n_e(v)$.

The choice of threshold turns distance between logarithmic functions into bit disagreement. For independent uniform edges e, e', interchange of nonnegative sums and integrals and (4.3) give

$$2\mathbb{E}_{e,e'} h(d_e/d_{e'}) = \mathbb{E}_{e,e'} \sum_v^* |n_e(v) - n_{e'}(v)| = 2 \int \varrho(1 - \varrho) \leq 2S. \tag{4.9}$$

4.3 Excluding a bounded minority-bit integral

Proposition 4.3 (The bounded alternative is impossible)**.** *For every sequence of incidence subgraphs in Proposition 2.1, the quantity S defined in (4.8) satisfies $S \to \infty$.*

We first use matching labels to transfer height control from edges to points. The argument is useful because it bounds heights of differences before any translation has been chosen.

Lemma 4.4 (Transferring edge heights)**.** *Suppose a probability law ω on two finite vertex sets $P \times Q$, with point marginal μ, has the following property: for all assignments of finite labels to its vertices, two independent μ-points disagree with probability at most C_0 times the edge disagreement probability. For any coordinate $U : P \cup Q \to \mathcal{N}$,*

$$\mathbb{E}_{p,p' \sim \mu} h(U_p - U_{p'}) \le C_1 \big(\mathbb{E}_{(p,q)\sim\omega} h(U_p - U_q) + 1 \big),$$

where C_1 depends only on C_0.

Proof. Fix an entry v and $s \ge 0$. At a nonarchimedean entry give two vertices the same label if their U-coordinates differ by at most e^s in absolute value. The strong triangle inequality makes this an equivalence relation. At an archimedean entry, use the cell of a square grid in $\mathbb{C}$ of side e^s, shifted by a uniformly random fractional vector. Only the labels occurring on the finite vertex set matter.

For complex numbers whose distance is a, the probability of different grid cells is at most

$$\min(1, \sqrt{2}\, a e^{-s}).$$

Indeed a mismatch in either real coordinate has probability at most its coordinate distance divided by the side length, and the two coordinate distances sum to at most $\sqrt{2}a$. Its integral over $s \ge 0$ is at most $C(1 + \log^+ a)$. At a nonarchimedean entry the integral of mismatch is exactly $\log^+ |U_p - U_q|_v$, with value zero for a zero difference. Thus the integrated edge mismatch, averaged over shifts at infinity, is at most $C(\mathbb{E}_{(p,q)\sim\omega} h(U_p - U_q) + 1)$.

Apply the assumed label inequality at each fixed level and shift. For a point pair at a nonarchimedean entry, mismatch is exactly the event $|U_p - U_{p'}|_v > e^s$. At infinity, distance greater than $\sqrt{2}e^s$ forces mismatch for every shift. Integration of these tail events bounds the point-pair height, with an additional $\log \sqrt{2}$ from the total archimedean weight. This proves the result. □

Proof of Proposition 4.3. Suppose $S \le S_0$ on a subsequence. All constants in this proof may depend on the fixed S_0. By (4.9), there is an edge value d_0 such that

$$\mathbb{E}_E h(d_e/d_0) \le S_0.$$

Temporarily replace the two coordinates by z/d_0 and $d_0 w$. Their edge coordinate differences are $D_e = d_e/d_0$ and D_e^{-1}, whose mean heights are at most S_0. Lemma 2.5 supplies the label hypothesis of Lemma 4.4. Applying it to both coordinates bounds the sum of the two mean point-pair difference heights. Averaging over the second point then chooses $p_0 \in P$ such that the translated coordinates

$$Z_x = (z_x - z_{p_0})/d_0, \qquad W_x = d_0(w_x - w_{p_0})$$

satisfy a fixed bound on $\mathbb{E}_\mu(h(Z_p) + h(W_p))$. Since $Z_q = Z_p - D_{pq}$ and $W_q = W_p - D_{pq}^{-1}$ on an edge, Lemma 4.2 also bounds the center means. Thus there is a fixed $C_h < \infty$ with

$$\mathbb{E}_\mu(h(Z_p) + h(W_p)) + \mathbb{E}_\kappa(h(Z_q) + h(W_q)) + \mathbb{E}_E(h(D_e) + h(D_e^{-1})) \le C_h. \tag{4.10}$$

The edge identity is still $(Z_p - Z_q)(W_p - W_q) = 1$.

We next give the selection argument, keeping the embedding independent of the points it tests. For *any* fixed finite history map H on P, sample $(p, \widehat{p}, q_1, q_2, q_3)$ by the law (3.13) with three neighbors, where ω is the uniform law on E, so that $\omega(q \mid p) = \mathbf{1}_{\{(p,q)\in E\}}/d(p)$, and independently choose a uniformly random embedding σ. As recorded after (3.13), both p and $\widehat{p}$ have marginal μ; each q_j has marginal κ; and each (p, q_j) has the original edge law. In particular, the three coordinate differences $D_j = D_{pq_j}$ have the correct individual edge marginals. The height inequality bounds $\mathbb{E}h(Z_{\widehat{p}} - Z_p)$ and each $\mathbb{E}h(D_j - D_l)$ using only (4.10). Explicitly,

$$\mathbb{E}h(Z_{\widehat{p}} - Z_p) \le 2\mathbb{E}_\mu h(Z_p) + \log 2, \qquad \mathbb{E}h(D_j - D_l) \le 2\mathbb{E}_E h(D_e) + \log 2.$$

These bounds use the individual marginals, so they are uniform in H.

Choose a fixed $B \ge 2$ so large that, by (4.5), the probability that any of

$$Z_p, W_p, Z_{\widehat{p}}, W_{\widehat{p}}, (Z_{q_j}, W_{q_j}, D_j, D_j^{-1})_{j=1}^3$$

has embedded absolute value greater than B is less than $1/100$. Then choose a fixed $0 < r < 1$ so small that (4.6) makes the probability of any of

$$\begin{aligned} &p \ne \widehat{p}, \quad |\sigma(Z_{\widehat{p}} - Z_p)| < r, \\ &D_j \ne D_l, \quad |\sigma(D_j - D_l)| < r \qquad (1 \le j < l \le 3) \end{aligned}$$

less than $1/100$ in total. The equalities have deliberately been excluded from these small-difference events. These estimates hold for every H because they were integrated over a fresh independent embedding using only the fixed marginal bounds.

Choose now

$$0 < \tau < \min\left(1, \frac{r^4}{128B^5}\right). \tag{4.11}$$

At an embedding, encode each point by quantizing the real and imaginary parts of both Z_p, W_p in $[-B-1, B+1]$, in half-open intervals of length at most τ. Add an overflow code for points outside this box, using an endpoint convention to make a partition. This is a finite alphabet of size $M = M(B, \tau)$ independent of t and of the field degree. An acceptance test at q asks whether the point code is attained by some actual neighbor of q. The actual point on an edge always passes.

By Lemma 2.7, $D = o(\log t)$. Choose $K \to \infty$ with $K = o(\log t)$ and $D/K \to 0$. Apply Corollary 3.2 to independent uniform embeddings as tests. It gives a fixed strict-prefix history map with at most $M^{K-1} = t^{o(1)}$ values for which the predicted acceptance failure at a fresh embedding is $o(1)$. The choice of history is made after averaging the prediction failure over that fresh embedding; the embedding itself is not fixed. The geometric error bounds just proved remain valid for this selected history.

Use this H in the law (3.13) with three neighbors, as above: one common prediction $\widehat{p}$ and three independently sampled neighbors of p. Each pair $(\widehat{p}, q_j)$ has the prediction law in the acceptance estimate, so all three tests pass outside probability $o(1)$ by a union bound. Moreover, by (3.14),

$$\mathbb{P}(\widehat{p} = p) \le |H(P)| \max_p \mu(p) = t^{-1+o(1)}.$$

For $j \ne l$, the conditional uniform neighbor law gives $\mathbb{P}(q_j = q_l) \le 1/\min_p d(p) = o(1)$. Distinct centers have distinct Z-coordinates, so $q_j \ne q_l$ implies $D_j \ne D_l$. The degree and atom estimates

used here follow from Proposition 2.1. Combining all these estimates gives positive probability that all three tests pass, all displayed quantities are bounded by B, and

$$|\sigma(Z_{\widehat{p}} - Z_p)| \geq r, \qquad |\sigma(D_j - D_l)| \geq r \quad (j < l). \tag{4.12}$$

Fix such an outcome and suppress σ from the notation. Successful acceptance provides, for each j, a genuine neighbor of q_j in the same code as $\widehat{p}$. Since the latter has coordinates bounded by B, this is a nonoverflow code, and the two complex coordinate errors are at most $\sqrt{2}\tau$. Comparing with that neighbor's exact edge equation gives

$$|(Z_{\widehat{p}} - Z_{q_j})(W_{\widehat{p}} - W_{q_j}) - 1| \leq 8B\tau.$$

Indeed the two factors at $\widehat{p}$ have absolute value at most $2B$. Changing each by at most $\sqrt{2}\tau$ changes their product by at most $4\sqrt{2}B\tau + 2\tau^2 \leq 8B\tau$, since $B \geq 2$ and $\tau < 1$. Set $\Delta Z = Z_{\widehat{p}} - Z_p$ and $\Delta W = W_{\widehat{p}} - W_p$. Subtracting the exact equation for (p, q_j) yields

$$\Delta Z/D_j + \Delta W D_j + \Delta Z \Delta W = E_j, \qquad |E_j| \leq 8B\tau.$$

Subtract the equation with $j = 1$ from those with $j = 2, 3$. The coefficient matrix for $(\Delta Z, \Delta W)$ has determinant

$$\det \begin{pmatrix} D_2^{-1} - D_1^{-1} & D_2 - D_1 \\ D_3^{-1} - D_1^{-1} & D_3 - D_1 \end{pmatrix} = \frac{(D_1 - D_2)(D_1 - D_3)(D_2 - D_3)}{D_1 D_2 D_3}. \tag{4.13}$$

Its absolute value is at least r^3/B^3 by (4.12). Each matrix entry is at most $2B$, and each right-hand side is at most $16B\tau$. Cramer's rule therefore gives

$$|\Delta Z| \leq \frac{64B^5\tau}{r^3} < r/2,$$

contradicting (4.12). This excludes every bounded subsequence of S and proves the proposition. □

4.4 Coordinate states on the original levels

We now return to the original coordinates z, w, intervals I_v, and thresholds n_e. By Proposition 4.3, $S \to \infty$. At a level (v, s) give each vertex x a state

$$\mathbf{y}_x(v, s) = (A_x(v, s), B_{1,x}(v, s)).$$

At a nonarchimedean entry, A_x is the equivalence class for $|z_x - z_{x'}|_v \leq e^{-s}$, and $B_{1,x}$ is that for $|w_x - w_{x'}|_v \leq e^s$. At an archimedean entry associated with σ, A_x is the half-open square-grid cell containing $\sigma(z_x)$ at side length e^{-s}, and $B_{1,x}$ is the cell containing $\sigma(w_x)$ at side length e^s. For each of these two grids choose a fractional shift in $[0, 1)^2$ and use that same fractional shift, scaled with the side length, at every level of the entry.

For a state $y = (a, b)$, R_y is the set (3.15) of states that agree with y in exactly one coordinate. Call an edge *consistent* at a level if $Y_e = 1$ and exactly its A-labels agree, or if $Y_e = 0$ and exactly its B_1-labels agree. Thus consistency implies that the endpoint states agree in exactly one coordinate, and also specifies which coordinate agrees. Define

$$m(v, s) = \mathbb{P}_E\{e \text{ is inconsistent at } (v, s)\}, \qquad m_e = \int \mathbf{1}_{\{e \text{ is inconsistent}\}}. \tag{4.14}$$

Lemma 4.5 (State consistency)**.** *The fractional shifts can be fixed so that*

$$\int m = \mathbb{E}_E m_e = O(1), \tag{4.15}$$

with an absolute constant. At each nonarchimedean entry every edge is consistent outside finitely many threshold levels. If two vertices have matching A-labels, then

$$|z_x - z_{x'}|_v \le e^{-s} \text{ at nonarchimedean } v, \qquad |z_x - z_{x'}|_v \le \sqrt{2}\, e^{-s} \text{ at archimedean } v;$$

matching B_1-labels similarly implies $|w_x - w_{x'}|_v \le e^s$ at nonarchimedean entries and $|w_x - w_{x'}|_v \le \sqrt{2}\, e^s$ at archimedean entries. All the level functions can be chosen measurable.

Proof. At a nonarchimedean entry, $n_e > s$ means $|d_e|_v < e^{-s}$ and $|d_e^{-1}|_v > e^s$, so exactly A agrees. For $n_e < s$ the two conclusions reverse. Only $s = n_e$ is excluded, a null set for the level measure.

At infinity, put $r = |s - n_e|$. The smaller coordinate difference divided by its grid scale is e^{-r}, so its mismatch probability under a uniform shift is at most $\sqrt{2}e^{-r}$. The larger coordinate difference divided by its scale is e^r; it cannot lie in the same cell once $r \ge \log\sqrt{2}$, by the diameter of a half-open square. Consequently the expected inconsistency indicator is at most

$$\mathbf{1}_{\{|s-n_e|\le \log\sqrt{2}\}} + \sqrt{2}e^{-|s-n_e|}.$$

Its integral over the full real line is bounded by an absolute constant, and hence so is its integral over I_v. Sum over entries using total archimedean weight one, then average over edges. There are only finitely many archimedean shifts to choose at each index, so averaging fixes shifts satisfying (4.15).

The match implications follow directly from the equivalence classes and the square diameter. For measurability, label nonarchimedean classes by the first vertex in a fixed ordering of $P \cup Q$, and use integer cell indices at infinity. All class comparisons are threshold comparisons and all cell indices are measurable functions of s. □

We have obtained a space of tests of finite measure with integrated minority-bit probability $S \to \infty$, while the integrated discrepancy between an edge's threshold bit and its two coordinate agreements remains $O(1)$. This makes that discrepancy negligible relative to S.

5 Level decomposition and difference bounds

We now use $S \to \infty$, as guaranteed by Proposition 4.3. Our inputs are the expanding incidence graph, the threshold bits $Y_e(v,s) = \mathbf{1}_{\{n_e(v)>s\}}$, and the coordinate states $\mathbf{y}_x = (A_x, B_{1,x})$ of Lemma 4.5. Recall that an edge is consistent when precisely its A labels match if its bit is 1, and precisely its B_1 labels match if its bit is 0. The inconsistency probability $m(v,s)$ satisfies $\int m = O(1)$. The purpose of this section is to compare two-edge samples at a point or a center. Their integrated scores will have a small upper bound from the product formula and a lower bound recording exceptions to dominant coordinate labels. These bounds also hold after conditioning edge pairs at a point, defined below, on a short list of histories.

All level integrals use the weighted measure of Section 4.2. In particular $f = \min(\mathbb{P}_E(Y = 0), \mathbb{P}_E(Y = 1))$ and $S = \int f$. Write $d_{P,\min} = \min_{p\in P} d(p)$ and $d_{\min,Q} = \min_{q\in Q} d(q)$.

5.1 Parameters, levels with $f < \epsilon$, and histories

The parameters must make prediction errors small even after taking several samples, while keeping the number of possible histories small relative to a point degree.

Lemma 5.1 (Parameter choice)**.** *There are integers $K, L, k \to \infty$ such that*

$$K \log L = o(\log t), \qquad D/K = o(1), \tag{5.1}$$

$$kK \log L = o(\log t), \qquad k^{20}\left((D/K)^{1/4} + L^{-1} + S^{-1} + d_{\min,Q}^{-1}\right) = o(1). \tag{5.2}$$

Proof. The incidence estimates give $D = o(\log t)$ and $d_{\min,Q} \to \infty$, and $S \to \infty$ by Proposition 4.3. Take K and L as in Section 3.4; they satisfy (5.1). The sequence $\eta_t = (D/K)^{1/4} + L^{-1} + S^{-1} + d_{\min,Q}^{-1}$ is positive and tends to zero, so the integer k chosen there with this η_t satisfies (5.2). In particular $k = o(\log t)$ since $K \log L \to \infty$. □

Fix such K, L, k for the rest of the argument, and set $\epsilon = k^{-1/16}$ and $e_t = (D/K)^{1/4} + L^{-1}$. The hierarchy gives $k^3/S = o(1)$, $k^{3+1/16} e_t = o(1)$, and $k = t^{o(1)}$.

Partition the levels by minority-bit and inconsistency probabilities:

$$\mathcal{C} = \{f \geq \epsilon\}, \qquad \mathcal{B} = \{f < \epsilon,\ m > 1/k\}, \qquad \mathcal{G} = \{f < \epsilon,\ m \leq 1/k\}.$$

Their first two widths satisfy

$$W_c := \int \mathbf{1}_{\mathcal{C}} \leq S/\epsilon, \qquad W_b := \int \mathbf{1}_{\mathcal{B}} \leq k \int m = O(k) = o(S). \tag{5.3}$$

At each place, $\mathcal{C}$ is an interval, possibly empty or of zero length: the function $s \mapsto \mathbb{P}_E(n_e(v) > s)$ is nonincreasing.

At a level in $\mathcal{G}$ let $Y_* \in \{0, 1\}$ be the minority bit, which is unique for all sufficiently large t. On every consistent majority-bit edge the B_1 labels match if $Y_* = 1$, and the A labels match if $Y_* = 0$. Thus the mismatch probability of this coordinate is at most $f + m$. By Lemma 2.5 it has a label ℓ whose two marginal exception probabilities have sum $O(f + m)$. Fix one such label measurably, and let $I_x(v, s)$ indicate that the relevant coordinate of x differs from ℓ. Write

$$h_P(v, s) = \mathbb{E}_\mu I_p(v, s), \quad h_Q(v, s) = \mathbb{E}_\kappa I_q(v, s), \quad H_x^* = \int_{\mathcal{G}} I_x(v, s).$$

Here and below I_x is used only on $\mathcal{G}$. Fixed orderings of the finite vertex sets settle all choices of labels, so measurability causes no additional restriction.

Lemma 5.2 (Exception mass and joint exceptions)**.** *Uniformly on $\mathcal{G}$,*

$$h_P + h_Q \leq C(f + m) = o(1), \qquad f \leq h_P + h_Q + m. \tag{5.4}$$

Consequently

$$\mathbb{E}_\mu H_p^* + \mathbb{E}_\kappa H_q^* = O(S). \tag{5.5}$$

Furthermore, with (p, q, r) distributed by the law of an edge pair at a center (2.11),

$$\int_{\mathcal{G}} \mathbb{P}_E(I_p = I_q = 1) = o(S), \qquad \int_{\mathcal{G}} \mathbb{P}(I_p = I_r = 1) = o(S). \tag{5.6}$$

Proof. The first inequality was just proved, and its right side is uniformly $O(\epsilon+1/k) = o(1)$. If both endpoints have the dominant label, a consistent edge has the majority bit. Every minority-bit edge therefore has an exceptional endpoint or is inconsistent, proving the second inequality. Integrating and using $\int m = O(1)$ proves (5.5) because $S \to \infty$.

For the joint errors, apply Lemma 2.6 with the uniform marginal bound $\delta_t = C(\epsilon + k^{-1}) \to 0$ from (5.4). Its edge estimate gives $\mathbb{P}_E(I_p = I_q = 1) = o(1)(h_P + h_Q)$, and its estimate for edge pairs at a center (2.14) gives $\mathbb{P}(I_p = I_r = 1) = o(1)h_P$. These are exactly the laws of edges and of edge pairs at a center used here. Both multipliers are uniform over all levels in $\mathcal{G}$, so they can be taken outside the integrals. Equation (5.5) then gives (5.6). □

We next apply the prediction device only to levels in $\mathcal{C}$. When $W_c > 0$, sample a fresh level from their normalized measure and use Corollary 3.3. An inconsistent edge accounts for every failure of agreement in exactly one true coordinate. We obtain a fixed history map H with $|H(P)| \le L^{2K}$ such that, writing ν_h for the true-state law of a point of marginal μ conditional on history h at the fresh level,

$$k^3 \int_{\mathcal{C}} \mathbb{E}_E[1 - \nu_{H(p)}(R_{\mathbf{y}_q})] = o(S). \tag{5.7}$$

Indeed the integral before multiplication by k^3 is at most $\int_{\mathcal{C}} m + O(W_c e_t)$, and Lemma 5.1 and (5.3) apply. If $W_c = 0$, take the empty history, and the assertion is vacuous.

For each center q, a *history list* is an ordered list $\theta = (h_1, \dots, h_k)$ obtained by sampling k independent neighbors of q and recording their histories. Let $a(q, \theta) = \mathbb{P}(\theta \mid q)$. The number N_θ of possible history lists satisfies

$$N_\theta \le L^{2Kk} = t^{o(1)}. \tag{5.8}$$

The history map is already fixed. Thus $a(q, \theta)$ uses no fresh level. In the joint law

$$\widetilde{\omega}(p, q, \theta) = \frac{\mathbf{1}_{\{(p,q)\in E\}}}{|E|} a(q, \theta), \tag{5.9}$$

the point p and the history list are independent conditional on q.

5.2 Three fixed pair experiments

We use three ways to sample a base and two opposite endpoints:

(i) Q: sample $q \sim \kappa$, then p, r independently from the conditional edge law at q;

(ii) P: sample $p \sim \mu$, then q, q' independently from the conditional edge law at p;

(iii) $P\theta$: sample (p, θ) from its marginal under $\widetilde{\omega}$, then q, q' independently from $\widetilde{\omega}(q \mid p, \theta)$.

Only positive-mass bases are used. Each sample is an edge pair at a center in experiment Q, an edge pair at a point in experiment P, or an edge pair at a point conditioned on a history list in experiment $P\theta$. The Q experiment has the law (2.11). In each experiment write x for the base vertex, y_1, y_2 for its opposite endpoints, and Y_1, Y_2 for the corresponding edge bits. The base in the third experiment includes θ. At a level let $\rho = \mathbb{P}(Y_1 = 1 \mid \text{base})$ under the full indicated law. These laws are fixed across all levels. Define

$$C_i = \int_{\mathcal{C}} \mathbb{E}_{\text{base}}[2\rho(1-\rho)], \qquad i \in \{Q, P, P\theta\}. \tag{5.10}$$

Lemma 5.3 (Marginals and diagonal pairs)**.** *In each of the Q, P and $P\theta$ experiments, either edge slot has the uniform law on E after averaging over the base. In the $P\theta$ experiment, for every fixed $p \in P$, averaging the conditional probability of $q = q'$ over θ given p gives at most $N_\theta/d(p)$.*

Proof. The first assertion is immediate for the Q and P edge pairs. For the third, the (p, q) marginal of (5.9) is the edge law since $\sum_\theta a(q, \theta) = 1$; sampling a conditional slot preserves this marginal. More explicitly, a slice of positive probability $w_\theta = \sum_{p,q} \widetilde{\omega}(p, q, \theta)$ has one-edge law $\widetilde{\omega}(p, q, \theta)/w_\theta$ and must be averaged with weight w_θ. One can also attach a history list independently at a Q base: conditional on (q, θ) its two point neighbors still have their original law, and its single-slot law in each slice is the same as for $P\theta$.

Fix p and put $a_p(\theta) = \mathbb{P}(\theta \mid p)$. For $a_p(\theta) > 0$,

$$\widetilde{\omega}(q \mid p, \theta) = \frac{\mathbf{1}_{\{(p,q)\in E\}} a(q, \theta)}{d(p) a_p(\theta)} \le \frac{1}{d(p) a_p(\theta)}.$$

Its collision probability is at most its largest atom. Multiplying by $a_p(\theta)$ and summing over at most N_θ positive-mass history lists gives the asserted bound pointwise in p. □

We now define scores that reward matching opposite endpoints and charge differing bits. If $Y_1 = Y_2 = 1$, let N be the event that the B_1 labels of y_1, y_2 match; if $Y_1 = Y_2 = 0$, let N instead mean that their A labels match; set N to be false when the bits differ. Thus N concerns the coordinate which a consistent edge of that bit does not share with its base. For independent bits with parameter ρ, equality has probability $\rho^2 + (1 - \rho)^2$, whereas disagreement has probability $2\rho(1 - \rho)$. The former is larger except at $\rho = 1/2$. Thus, when equal bits force N, a reward slightly above one near balance and slightly below one elsewhere can give a positive expected score. The decrease away from balance also makes room for the increase in the integrated upper bound below. Section 6 will identify the configurations in which equal bits force the required match. Set

$$\delta = \frac{1}{100}, \qquad \varpi = \frac{\delta^2}{2}, \qquad c(\rho) = \begin{cases} 1 + \varpi, & |\rho - \frac{1}{2}| \le \delta, \\ 1 - \varpi, & |\rho - \frac{1}{2}| > \delta. \end{cases}$$

At a level in $\mathcal{G}$ Y_* is the minority bit. At every level we charge one for bit disagreement and add the indicated reward:

$$\text{score} = -\mathbf{1}_{\{Y_1 \ne Y_2\}} + \begin{cases} 2I_x \mathbf{1}_{\{Y_1 = Y_2 = Y_*,\, N\}}, & \text{at a level in } \mathcal{G}, \\ c(\rho) \mathbf{1}_{\{Y_1 = Y_2,\, N\}}, & \text{at a level in } \mathcal{C}, \\ 0, & \text{at a level in } \mathcal{B}. \end{cases}$$

Let $\mathcal{D}_i$ be the expectation of this score integrated over all levels in experiment i.

5.3 The upper score bound

Proposition 5.4 (Upper score bound)**.** *The integrated scores of the three fixed pair experiments satisfy*

$$\mathcal{D}_i \le o(S) + 10\delta^2 \varpi C_i \qquad (i \in \{Q, P, P\theta\}). \tag{5.11}$$

The next estimate is the placewise mechanism behind the score bound. It identifies exactly when the coefficient $1 + \varpi$ can spend more than the available interval length.

Lemma 5.5 (Two intervals and a third interval)**.** *Let $b \le b'$, $T \ge 0$, and J be an interval of finite length, with any choice of included endpoints. Put $I_- = [b-T, b]$, $I_+ = [b', b'+T]$, and $a = |J \cap I_-| + |J \cap I_+|$, where $|\cdot|$ denotes length. For $0 < \lambda < 1$,*

$$\big[(1-\lambda)2T + 2\lambda a - 2T\big]_+ \le \lambda |J|\, \mathbf{1}_{\{b,b' \in \operatorname{int} J\}}. \tag{5.12}$$

If X has any probability law on $\mathbb{R}$, and $|\mathbb{P}(X > s) - 1/2| \le \delta$ throughout J, then

$$\mathbb{P}(X \in \operatorname{int} J) \le 2\delta. \tag{5.13}$$

Proof. The left side of (5.12) is $[2\lambda(a-T)]_+$. For it to be positive both intersections must have positive length, since either has length at most T. As J is an interval, this puts both b, b' in its open interior, also when $b = b'$. Moreover $a \le 2T$ and $a \le |J|$, since I_- and I_+ have disjoint interiors. Hence $2(a-T) \le a \le |J|$. For (5.13), take $u < v$ in the interior. Then

$$\mathbb{P}(u < X \le v) = \mathbb{P}(X > u) - \mathbb{P}(X > v) \le 2\delta.$$

Exhausting the open interior by such intervals proves the claim. The empty-interior case is immediate. □

Figure 1 illustrates why positive excess in Lemma 5.5 requires both thresholds to lie inside the interval.

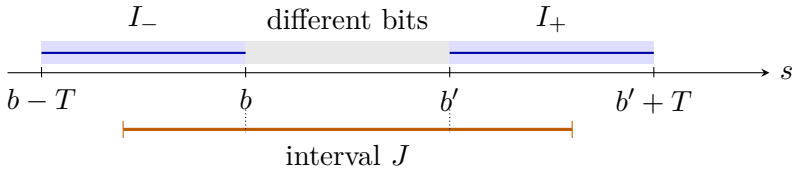


Figure 1: The intervals $I_- = [b-T, b]$ and $I_+ = [b', b'+T]$, with $b \le b'$ and $T \ge 0$, flank the bit-difference interval. The interval J in Lemma 5.5 creates excess only if its total overlap with the two flanks exceeds T; then both thresholds lie strictly inside J.

Proof of Proposition 5.4. First fix a base and two distinct opposite endpoints, and write d, d' for their incident coordinate differences. Distinct real endpoints have distinct z coordinates, so $d \ne d'$. At a place v let

$$b = \min(n_e(v), n_{e'}(v)), \quad b' = \max(n_e(v), n_{e'}(v)), \quad u = -\log|d-d'|_v, \quad \alpha = u - b.$$

The two opposite-endpoint coordinate differences are $d - d'$ and $1/d - 1/d'$ up to signs. Their logarithms satisfy

$$\log|d-d'|_v = -b - \alpha, \qquad \log|1/d - 1/d'|_v = -u + b + b' = b' - \alpha,$$

because $|1/d - 1/d'|_v = |d-d'|_v/(|d|_v|d'|_v)$. Therefore

$$-\log|(d-d')(1/d - 1/d')|_v = 2\alpha - (b'-b). \tag{5.14}$$

All thresholds lie within the chosen level interval, so the integrated bit-difference charge at this place is exactly $b' - b$.

Let $c_v = \log 2$ at archimedean entries and $c_v = 0$ otherwise. The triangle inequality gives $\alpha \geq -c_v$, hence $T_v := \alpha + c_v \geq 0$. By Lemma 4.5, equality of a grid label at infinity bounds the corresponding difference by $\sqrt{2}$ times the grid scale, and at a finite place a matching label bounds it by the scale itself; in both cases the factor is at most e^{c_v}, since $\sqrt{2} < 2$. If both bits are 1, a positive charge requires $s < b$ and a B_1 match, which gives

$$s \geq b' - \alpha - c_v \geq b - T_v.$$

If both bits are 0, the analogous A match gives $s \geq b'$ and $s \leq b + \alpha + c_v \leq b' + T_v$. Thus all positive charges lie in

$$I_- = [b - T_v, b], \qquad I_+ = [b', b' + T_v]. \tag{5.15}$$

Enlarging to these full intervals is legitimate even if part of them lies outside the chosen level interval.

Suppose a positive charge on $\mathcal{G}$ is counted with minority bit 1 at a threshold s_0. Then $s_0 < b$ and $\mathbb{P}_E(n_e(v) > s_0) < \epsilon$. By monotonicity every level in $\mathcal{C}$ and every level with minority bit 0 lies below s_0; none can produce a positive charge in I_+. All minority-1 positive charges lie in I_- by definition. The case of a positive charge on $\mathcal{G}$ with minority bit 0 is symmetric. Thus, if any positive charge on $\mathcal{G}$ is counted, all positive charges are on one side, their coefficients are at most 2, and their total is at most $2T_v$.

Otherwise only positive charges on $\mathcal{C}$ remain. For this fixed base put

$$J_v = \{s : (v, s) \in \mathcal{C},\ |\rho(v, s) - 1/2| \leq \delta\}.$$

This is an interval because the conditional edge law is fixed across levels and both restrictions are intervals. Extend the coefficient $1 - \varpi$ across both full intervals in (5.15), increasing it by 2ϖ on J_v. Lemma 5.5 bounds the excess over $2T_v$ by

$$\varpi |J_v|\, \mathbf{1}_{\{n_e(v), n_{e'}(v) \in \operatorname{int} J_v\}}.$$

This expression is also a valid excess bound in the case of positive charges on $\mathcal{G}$, when no excess is needed. Combining it with the negative charge and (5.14), then summing over places, cancels the logarithm by the product formula for the fixed nonzero element $(d - d')(1/d - 1/d')$. The additive allowance is $2 \sum_v^* c_v = 2 \log 2$. Places of zero level width can be included in this sum: their positive and negative charges vanish and the same upper bound still holds.

Now average this bound over the fixed-base independent pair law, restricting to distinct pairs. Dropping that restriction only in the nonnegative excess term, (5.13) bounds its expectation by $4\delta^2 \varpi |J_v|$. Boundary atoms are not counted; distinct coordinate differences with equal thresholds cause no exception to this argument. On J_v, $2\rho(1 - \rho) \geq 2(1/4 - \delta^2)$. Averaging over bases and places therefore bounds the distinct-pair contribution by

$$O(1) + \frac{4\delta^2 \varpi}{2(1/4 - \delta^2)} C_i \leq O(1) + 10\delta^2 \varpi C_i.$$

It remains to bound equal endpoints, to which the product formula was not applied. Their positive score is at most $2(W_c + H_x^*)$ for a base vertex x, and their negative score is zero. The ordinary collision probabilities are $1/d(q)$ and $1/d(p)$. For edge pairs conditioned on a history list Lemma 5.3 gives the bound $N_\theta / d(p)$ pointwise in p. Consequently their total contributions are at most, respectively,

$$\frac{2}{d_{\min,Q}}(W_c + \mathbb{E}_\kappa H_q^*), \quad \frac{2}{d_{P,\min}}(W_c + \mathbb{E}_\mu H_p^*), \quad \frac{2N_\theta}{d_{P,\min}}(W_c + \mathbb{E}_\mu H_p^*).$$

Each is $o(S)$ by Lemmas 5.1 and 5.2, (5.8), and $d_{P,\min} = t^{1+o(a)}$. In particular multiplication by an unbounded H_p^* requires only its first moment, since the collision estimate holds for every p. Finally $O(1) = o(S)$. □

5.4 The lower bound outside $\mathcal{C}$

The upper bound is insensitive to whether the base lies at a point or a center. The next lower bound keeps track of that orientation. Its two main terms will cancel when one score for an edge pair at a point is added to one score for an edge pair at a center.

Proposition 5.6 (Lower bound outside $\mathcal{C}$)**.** *For each of the three pair experiments, let x be its base vertex and y an opposite endpoint. Its score integrated outside $\mathcal{C}$ is at least*

$$2\mathbb{E}H_x^* - 2\mathbb{E}H_y^* - o(S), \tag{5.16}$$

where both expectations use the original edge marginals on the appropriate sides. In particular this is $2\mathbb{E}_\kappa H_q^* - 2\mathbb{E}_\mu H_p^* - o(S)$ for Q, and its opposite main term for either P or $P\theta$.

Proof. At a level in $\mathcal{G}$ denote the inconsistency indicator in slot j by M_j. If $I_x = 1$, both other endpoints have the dominant label, and both edges are consistent, their dominant-coordinate labels match each other but differ from the base's. Both bits must therefore be the minority bit and the required nonshared coordinate matches. The positive score is consequently at least

$$2I_x - 2I_xI_{y_1} - 2I_xI_{y_2} - 2M_1 - 2M_2.$$

If both opposite endpoints have the dominant label and both edges consistent, their bits agree regardless of whether the base is exceptional. Hence the negative charge is at most $I_{y_1}+I_{y_2}+M_1+M_2$. Average these two pointwise bounds. Each slot has the original edge marginal by Lemma 5.3, even with history lists. The integrals of the joint terms $I_xI_{y_j}$ are $o(S)$ by (5.6); the inconsistency integrals are $O(1)$. This gives (5.16) on $\mathcal{G}$. On $\mathcal{B}$ there are no positives and the negative score is at least -1, whose total cost is $W_b = o(S)$. □

6 Levels in $\mathcal{C}$ and positive exception mass on $\mathcal{G}$

We use the histories, history lists and three edge-pair scores constructed in Section 5. The first objective is to show that levels in $\mathcal{C}$ contribute only $o(S)$ bit variation at a center. The second is to show that the point exceptions on levels in $\mathcal{G}$ retain a positive fraction of S. Both conclusions come from comparing the upper and lower edge-pair bounds, but the comparison first requires a precise description of the states permitted by a history list.

Throughout this section, $S \to \infty$, the parameters satisfy Lemma 5.1, and H is the fixed history map from Equation (5.7). At a level, ν_h denotes the distribution of the true state (A, B_1) of a point sampled from μ conditional on $H = h$. Dependence on the level is suppressed. Recall from (3.15) that the states agreeing with $y = (a, b)$ in exactly one coordinate form

$$R_y = \{(u, v) : (u = a,\ v \neq b) \text{ or } (u \neq a,\ v = b)\}.$$

A row means a set with fixed first coordinate; a column has fixed second coordinate.

Proposition 6.1 (Bit disagreement on $\mathcal{C}$)**.** *Under the standing parameter and history choices, the bit-disagreement integrals for experiments Q and $P\theta$ from (5.10) satisfy*

$$C_Q + C_{P\theta} = o(S). \tag{6.1}$$

Moreover their exception means on $\mathcal{G}$ satisfy

$$\mathbb{E}_\kappa H_q^* \le \mathbb{E}_\mu H_p^* + o(S). \tag{6.2}$$

Proposition 6.2 (Positive exception mass on $\mathcal{G}$)**.** *For the same incidence subgraphs, parameters and fixed history, the point exceptions on $\mathcal{G}$ satisfy*

$$\liminf_{t\to\infty} \frac{\mathbb{E}_\mu H_p^*}{S} > 0. \tag{6.3}$$

The proof of Proposition 6.1 is in Section 6.2. On levels in $\mathcal{C}$ the state classification gives a lower bound with coefficient $\varpi/4$ for the sum of the two variations. The terms from outside $\mathcal{C}$ cancel between the two orientations by Proposition 5.6, whereas Proposition 5.4 bounds the same sum with coefficient $10\delta^2\varpi < \varpi/4$. The remaining proposition is proved in Section 6.3: if the point-exception mean were $o(S)$, a second comparison would also give $C_P = o(S)$, and the matching-label estimate would force $S = o(S)$.

6.1 From sampled histories to the actual point

For a center q, define the mixture of predictions

$$\widetilde{\nu}_q = \mathbb{E}_{p|q}\nu_{H(p)}.$$

For its history list $\theta = (h_1, \dots, h_k)$, define the finite set of candidate center states

$$\mathcal{Y}_\theta = \{y \text{ a center state} : \nu_{h_j}(R_y) \ge 1 - k^{-2} \text{ for every } 1 \le j \le k\}. \tag{6.4}$$

The actual state $\mathbf{y}_q$ is usually a candidate: the union bound and Markov's inequality give, after integration on $\mathcal{C}$,

$$\int_{\mathcal{C}} \mathbb{P}(\mathbf{y}_q \notin \mathcal{Y}_\theta) \le k^3 \int_{\mathcal{C}} \mathbb{E}_E[1 - \nu_{H(p)}(R_{\mathbf{y}_q})] = o(S). \tag{6.5}$$

We also need R_y for every candidate y to have large mass under $\widetilde{\nu}_q$. The following elementary sampling bound supplies this uniformity. For its first assertion, we specialize the ε-net double-sampling argument to the sets R_y [14, Lemmas 3.4–3.5]. The trace bound for these sets, the weighted-law formulation, and the subsequent transfer of concentration of the conditional law are established here.

Lemma 6.3 (Sampling the sets R_y)**.** *Let ν be a probability measure on a finite product set, let $k \ge 1$ be an integer, and let $X_1, \dots, X_k$ be independent samples from ν. For $0 < \alpha < 1$ and $k\alpha \ge 8$,*

$$\mathbb{P}(\exists y : \nu(R_y) < 1 - \alpha,\ X_1, \dots, X_k \in R_y) \le 2(2k+1)^2\, 2^{-k\alpha/2}.$$

Consequently, when k is sufficiently large, the history lists in (6.4) satisfy

$$\mathbb{P}(\exists y \in \mathcal{Y}_\theta : \widetilde{\nu}_q(R_y) < 1 - k^{-1/4} \mid q) \le 4(2k+1)^2\, 2^{-k^{3/4}/2} \tag{6.6}$$

at each level, uniformly in q.

Proof. Adjoin independent samples $X_{k+1}, \dots, X_{2k}$. If the first k samples admit a set R_y as in the event, select one by a fixed ordering. For this selected set R_y, the number of misses among the new samples has mean $k\beta$, where $\beta = \nu(R_y^c) > \alpha$, and variance at most $k\beta$. Thus the probability of fewer than $k\alpha/2$ misses is at most $4/(k\beta) \le 1/2$. It suffices to bound twice the probability that some R_y contains the first half and misses at least $k\alpha/2$ of the second half.

The joint sample mass is $\prod_{i=1}^{2k} \nu(X_i)$, which is unchanged by every permutation of the indices, even when ν is not uniform. We may therefore condition on the pooled $2k$ samples and split their indices uniformly into two halves. A set R_y induces at most $(2k+1)^2$ different subsets of these indices: its first coordinate can agree with one of at most $2k$ observed coordinates, or none, and the same holds for its second coordinate. If a pattern has m misses, the probability that they all fall in the second half is zero for $m > k$, and otherwise is

$$\frac{k(k-1)\cdots(k-m+1)}{(2k)(2k-1)\cdots(2k-m+1)} \le 2^{-m}.$$

The union bound proves the first assertion. Repeated sampled states cause no problem, since the argument concerns the indexed samples.

For the second assertion, after sampling the histories independently given q, independently sample a state from each corresponding ν_{h_j}. Before the histories are revealed these states are independent with law $\widetilde{\nu}_q$. If a candidate violating the required conclusion exists, select one as a function of the history list. Conditional on that history list, all the synthetic states lie in its set R_y with probability at least $1 - k \cdot k^{-2} = 1 - 1/k$. Apply the first assertion with $\alpha = k^{-1/4}$ and divide by $1 - 1/k$. □

Since $W_c \le k^{1/16} S$, the integral of the failure probability in (6.6) is $o(S)$. A candidate therefore constrains the mixture of predicted states. Concentration of this mixture does not by itself control the actual point: a set chosen from (q, θ) may use information about the point that its history does not contain. For rows, columns and sets of at most two states, the next lemma transfers this concentration to the actual point using exceptional sets determined only by the history and level.

Lemma 6.4 (Transfer for restricted sets)**.** *Fix a level and the joint edge–history list law*

$$\mathbb{P}(p, q, \theta) = \mathbb{P}_E(p, q)\mathbb{P}(\theta \mid q).$$

Let $0 < \eta < 1/2$. As a function of (q, θ), select a set $U(q, \theta)$ that is a row, a column, or a set of at most two states. On an event G determined by (q, θ), suppose

$$\widetilde{\nu}_q(U(q, \theta)) \ge 1 - 3\eta^2.$$

Then the actual point state satisfies

$$\mathbb{P}(G,\ \mathbf{y}_p \notin U(q, \theta)) \le 8\eta.$$

The set type and its coordinates may depend on both q and θ.

Proof. Conditional on (q, θ), the history of the actual point has its ordinary conditional law given q, because the history list is independent of the point given q. Hence on G,

$$\mathbb{E}[\nu_{H(p)}(U(q, \theta)^c) \mid q, \theta] \le 3\eta^2.$$

The event that this set has $\nu_{H(p)}$-mass less than $1 - \eta$ therefore has probability at most 3η.

For each history h, define three exceptional sets using only ν_h. If a row has mass at least $1-\eta$, it is unique, and the first exceptional set is its complement. If no such row exists, the first exceptional set is empty. Define the second set in the same way for columns. Their masses are at most η each.

If a set of at most two states has mass at least $1-\eta$, put

$$M_h = \{x : \nu_h(x) > \eta\}$$

and take M_h^c as the third exceptional set. Every set U of at most two states and mass at least $1-\eta$ contains M_h: omitting an atom of mass greater than η would violate that mass bound. Choosing one such U also gives

$$\nu_h(M_h^c) \le \nu_h(U^c) + \nu_h(U \setminus M_h) \le \eta + 2\eta = 3\eta.$$

If no such U exists, the third exceptional set is empty. The union of the three exceptional sets thus has mass at most 5η.

The actual point has marginal law μ, under which its state conditional on $H = h$ has distribution exactly ν_h. Consequently the probability of this union of exceptional sets is at most 5η under the entire edge–history list law. Outside it and the earlier nonconcentration event, the actual state belongs to whichever permitted set was selected. Adding the two error bounds proves the result. □

In the application, at most three candidates will be chosen by a fixed ordering at each level and history list. Outside the failure event bounded in (6.6), the intersection of their sets R_y has mixture mass at least $1 - 3k^{-1/4}$. The set will be this intersection or a row or column containing it. Its choice, the transfer event and its cost are specified with the bad edges in Section 6.2.

6.2 Classification and suppression of bit disagreement on $\mathcal{C}$

We now apply this transfer to the intersections of the sets R_y. Two distinct center states in one row, say (a, b) and (a, b'), give the intersection

$$R_{(a,b)} \cap R_{(a,b')} = \{(a, v) : v \notin \{b, b'\}\}. \tag{6.7}$$

A third center state outside that row cuts this intersection down to at most one point. The corresponding assertions hold for columns. If two states (a, b) and (a', b') differ in both coordinates, the intersection of their sets is

$$R_{(a,b)} \cap R_{(a',b')} = \{(a, b'), (a', b)\}. \tag{6.8}$$

Three states whose first coordinates are all distinct and whose second coordinates are all distinct have empty intersection of their sets R_y. Indeed each of the two points in (6.8) fails to share a coordinate with the third state. These observations give an exhaustive classification: a candidate set is empty, is a singleton, contains two distinct states sharing a coordinate, or consists of states with pairwise distinct coordinates. In the shared-coordinate case, either all candidates lie in that row or column, or a third lies outside it. In the pairwise-distinct case there are either two candidates or at least three.

Figure 2 shows the two-corner case: either endpoint state together with the bit determines the state at the other endpoint.

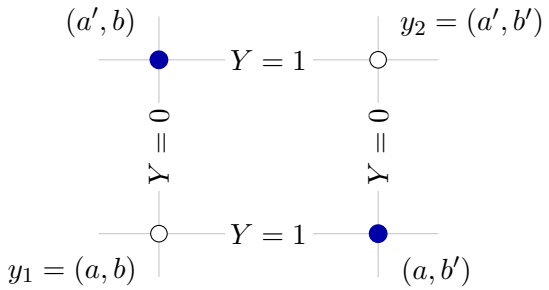


Figure 2: Two candidate center states differing in both coordinates admit exactly two point states in the intersection of their sets R_y. The filled points are those opposite corners. Within this four-state configuration, the bit of a consistent edge and the state at either endpoint determine the state at the other endpoint. Rows have fixed first label and columns have fixed second label. The diagram displays coordinate equalities of states.

For each history list with at least two candidates, choose witnesses by fixed orderings: two states sharing a coordinate, with a third outside that row or column when one exists; otherwise two candidates if there are exactly two, and three if there are more. If all candidates lie in the row or column of the shared-coordinate pair, let U_θ be that row or column. In the other cases let U_θ be the intersection of the sets R_y for the witnesses, which has at most two states. Set $U_\theta = \varnothing$ for empty or singleton candidate sets.

Let G be the event, determined by (q, θ) at this level, that $|\mathcal{Y}_\theta| \geq 2$ and $\widetilde{\nu}_q(R_y) \geq 1 - k^{-1/4}$ for every $y \in \mathcal{Y}_\theta$. On G, the at most three witness sets R_y give $\widetilde{\nu}_q(U_\theta) \geq 1 - 3k^{-1/4}$. Thus Lemma 6.4, with $\eta = k^{-1/8}$, gives

$$\mathbb{P}(G,\ \mathbf{y}_p \notin U_\theta) \leq 8k^{-1/8}.$$

Declare (p, q, θ) bad if its edge is inconsistent, if $\mathbf{y}_q \notin \mathcal{Y}_\theta$, if the failure event in (6.6) occurs, or if G holds and $\mathbf{y}_p \notin U_\theta$. Empty candidate sets force the second condition. For large k, empty witness intersections force the third, since their mixture mass cannot be at least $1 - 3k^{-1/4} > 0$; hence they have no good edges either. The transfer event contributes at most $8k^{-1/8}W_c \leq 8k^{-1/16}S = o(S)$. Together with the preceding estimates and the integrated inconsistency bound, this gives

$$\int_{\mathcal{C}} \mathbb{P}(p, q, \theta \text{ is bad}) = o(S). \tag{6.9}$$

The comparisons below must use the full conditional bit probability ρ in the definition of the scores. We never remove bad edges and renormalize that law. In particular the discontinuity of $c(\rho)$ at $1/2 \pm \delta$ introduces no error.

Here is the numerical estimate used in the two-corner case. For every $0 \leq \rho \leq 1$, with $\delta = 0.01$, $\varpi = \delta^2/2$, and c as in Section 5,

$$\begin{aligned} c(\rho)\big(\rho^2 + (1-\rho)^2\big) - 2\rho(1-\rho) &= (2\rho - 1)^2 + (c(\rho) - 1)\big(\rho^2 + (1-\rho)^2\big) \\ &\geq \varpi/2. \end{aligned} \tag{6.10}$$

For $|\rho - 1/2| \leq \delta$, discard the first square and use $\rho^2 + (1-\rho)^2 \geq 1/2$. Outside that interval the expression is at least $4\delta^2 - \varpi > \varpi/2$.

Proof of Proposition 6.1. Fix a level in $\mathcal{C}$. For each history list of positive probability, let $w_\theta = \mathbb{P}(\theta)$ and $b_\theta = \mathbb{P}(\text{bad edge} \mid \theta)$, with both probabilities taken under the joint law in Lemma 6.4. In the experiment Q, condition on θ and then sample q by its actual conditional law. The two points are still independent with their ordinary law given q. In the experiment $P\theta$, sample p conditional on θ, then two independent centers given (p, θ). Each slot in either experiment has the same conditional single-edge law $\mathbb{P}(p, q \mid \theta)$. Therefore a pair with at least one bad slot has probability at most $2b_\theta$ in either experiment.

Write $V_Q(\theta)$ and $V_{P\theta}(\theta)$ for the corresponding averages of $2\rho(1-\rho)$ over bases in this slice, and write $d_Q(\theta), d_{P\theta}(\theta)$ for its two expected scores on $\mathcal{C}$. Each V is at most $1/2$, and either score is at least the negative of its variation. We prove

$$d_Q(\theta) + d_{P\theta}(\theta) \geq \frac{\varpi}{4}\big(V_Q(\theta) + V_{P\theta}(\theta)\big) - Cb_\theta \tag{6.11}$$

with an absolute constant C. All coefficients in a score on $\mathcal{C}$ are bounded by $1 + \varpi$, so charging a pair with a bad slot always costs at most an absolute multiple of its probability.

An empty candidate set. Every edge is bad, so $b_\theta = 1$, and the boundedness of the scores proves (6.11) on increasing C.

One candidate state. At a fixed base (p, θ), two good edges have the same center state. Consistency forces their bits to agree, and the nonshared coordinate of their center states matches. Thus this edge-pair score is at least $1 - \varpi - O(b_\theta)$ after averaging over its bases. The score for an edge pair at a center is at least $-1/2$. Their sum is at least $1/2 - \varpi - O(b_\theta)$, which implies (6.11) because the two variations sum to at most one.

Two candidates sharing a coordinate, and all candidates in that row or column. The transfer using (6.7) places every good point in the same row or column. If it is a row, every good edge has matching A-coordinates and hence bit 1, by consistency; in the column case every good edge has bit 0. A bit difference in either pair experiment therefore requires a bad slot. This gives $V_Q(\theta) + V_{P\theta}(\theta) \leq 4b_\theta$. The lower score bound $d_i \geq -V_i$ now proves (6.11).

Two candidates sharing a coordinate, and a third outside that row or column. The three witness sets R_y intersect in at most one state. If the intersection is empty, all edges are bad. Otherwise all good point states equal that one state. At a center base, two good edges consequently have equal bits and the required nonshared-coordinate match. Its score is at least $1 - \varpi - O(b_\theta)$, and the other score is at least $-1/2$. The same estimate as in the singleton case applies.

No coordinate shared by distinct candidates. If there are at least three candidates, their sets R_y have empty intersection and all edges are bad. Otherwise there are exactly two candidates. The good point states lie in their two opposite corners from (6.8). At either type of base, a good edge's bit determines the state at the other endpoint. Thus two good edges with equal bits have the required collision. If β_x is the conditional probability of a bad slot at the base, then under its full pair law

$$\mathbb{P}(Y_1 = Y_2, N \mid \text{base}) \geq \rho^2 + (1-\rho)^2 - 2\beta_x.$$

The bit-disagreement probability is still exactly $2\rho(1-\rho)$. Thus the expected score is at least the left side of (6.10) minus $2(1+\varpi)\beta_x$. Averaging β_x over either type of base gives b_θ. Hence each score is at least $\varpi/2 - O(b_\theta)$, which proves (6.11).

This completes the classification. Multiply (6.11) by its actual slice weight w_θ, sum, and integrate over levels in $\mathcal{C}$. The error is $C\int_{\mathcal{C}} \sum_\theta w_\theta b_\theta = o(S)$ by (6.9). This argument allows individual b_θ to be large; no inverse slice weight or number of history lists enters. The contributions on $\mathcal{C}$ to $\mathcal{D}_Q + \mathcal{D}_{P\theta}$ are therefore at least

$$\frac{\varpi}{4}(C_Q + C_{P\theta}) - o(S).$$

The lower bounds in Proposition 5.6 outside $\mathcal{C}$ cancel their main terms, because the two base vertices are on opposite sides. Hence

$$\frac{\varpi}{4}(C_Q + C_{P\theta}) - o(S) \le \mathcal{D}_Q + \mathcal{D}_{P\theta} \le 10\delta^2\varpi(C_Q + C_{P\theta}) + o(S),$$

where the last inequality is Proposition 5.4. Since $10\delta^2 < 1/4$, this proves (6.1).

Finally the contribution on $\mathcal{C}$ to $\mathcal{D}_Q$ is at least $-C_Q$. Proposition 5.6 and Proposition 5.4 therefore give

$$2\mathbb{E}_\kappa H_q^* - 2\mathbb{E}_\mu H_p^* - C_Q - o(S) \le \mathcal{D}_Q \le 10\delta^2\varpi C_Q + o(S).$$

Use $C_Q = o(S)$ to obtain (6.2). □

6.3 A second score forces positive exception mass on $\mathcal{G}$

We have suppressed bit disagreement on $\mathcal{C}$ at centers. To show that point exceptions remain on levels with $f < \epsilon$, suppose temporarily that their mean is $o(S)$. Proposition 6.1 then gives the same bound for center exceptions. We will show that bit disagreement on $\mathcal{C}$ is also small at edge pairs at a point. Once both variations for edge pairs at points and centers are small, expansion forces the edge bit itself to be almost constant at nearly every level, contradicting $S = \int f$.

Proof of Proposition 6.2. If the conclusion fails, restrict to a subsequence on which $\mathbb{E}_\mu H_p^* = o(S)$. By (6.2), also $\mathbb{E}_\kappa H_q^* = o(S)$.

For a probability measure ν on states, let ν_A, ν_{B_1} denote its coordinate marginals, with squared norms equal to the sums of squares of their atom masses. Define

$$T = \int_{\mathcal{C}} \mathbb{E}_\mu \left[\|(\nu_{H(p)})_A\|_2^2 + \|(\nu_{H(p)})_{B_1}\|_2^2 - 1 \right]. \tag{6.12}$$

These norms count coordinate matches between two independent predicted points with the same history. This correction is needed when the good neighbors of a point have two states in one row or column and a third state outside it, while all three sets R_y have nearly full prediction mass. Their intersection then concentrates $\nu_{H(p)}$ on a single state, making the integrand in (6.12) close to one. The score for an edge pair at a point alone has only the general lower bound $-1/2$ in this case; adding the two gives a positive lower bound. We will compare $\mathcal{D}_P + T$ with C_P using (6.14). To retain the upper bound under the present assumption on exception mass on $\mathcal{G}$, we first prove

$$T \le o(S). \tag{6.13}$$

Choose a history with its marginal probability, then two independent points p', p'' from μ conditional on that history. This is one fixed coupling throughout all levels, and each point marginal is μ. At a level, the expected value of

$$F(p', p'') = \mathbf{1}_{\{A_{p'} = A_{p''}\}} + \mathbf{1}_{\{B_{1,p'} = B_{1,p''}\}} - 1$$

is the integrand in (6.12).

For a distinct pair put $Z = z_{p'} - z_{p''}$ and $W = w_{p'} - w_{p''}$, both nonzero. At a place with level interval $[l, u]$, the endpoint conditions from (4.7) say $l \le -\log|Z|_v$ and $u \ge \log|W|_v$. Put $a = -\log|Z|_v$, $b = \log|W|_v$, and let υ_v be $\log\sqrt{2}$ at infinity and zero otherwise. An A-match requires $s \le a + \upsilon_v$; a B_1-match requires $s \ge b - \upsilon_v$. Since $a \ge l$ and $b \le u$, the two matching

indicators have integrals at most $\min(u, a+\upsilon_v) - l$ and $u - \max(l, b-\upsilon_v)$, respectively. Including the constant term -1 in F therefore gives

$$\begin{aligned}\int_l^u F(p', p'')\,\mathrm{d}s &\le [\min(u, a+\upsilon_v) - l] + [u - \max(l, b-\upsilon_v)] - (u-l)\\ &= \min(u, a+\upsilon_v) - \max(l, b-\upsilon_v)\\ &\le -\log|Z|_v - \log|W|_v + 2\upsilon_v.\end{aligned}$$

In particular, if $a+\upsilon_v < b-\upsilon_v$, then $F = -1$ throughout the gap $(a+\upsilon_v, b-\upsilon_v) \subseteq [l, u]$. Thus the estimate also covers disjoint windows, when the bound is negative. The product formula, and total archimedean weight one, show that the integral over all levels is at most an absolute constant.

At a level in $\mathcal{G}$ where neither point is exceptional, their dominant coordinate labels match, so $F \ge 0$. In general at such a level

$$F \ge -\mathbf{1}_{\{p' \text{ exceptional}\}} - \mathbf{1}_{\{p'' \text{ exceptional}\}}.$$

The contribution on levels in $\mathcal{B}$ is at least $-W_b$. Removing all levels outside $\mathcal{C}$ from the full integral consequently gives, for each distinct pair,

$$\int_{\mathcal{C}} F(p', p'') \le O(1) + H^*_{p'} + H^*_{p''} + W_b.$$

On the diagonal the integral on $\mathcal{C}$ is exactly W_c. The selected coupling is the pair law described after (3.14), so its diagonal probability satisfies

$$\mathbb{P}(p' = p'') \le |H(P)| \max_p \mu(p) = t^{-1+o(1)}.$$

Here $|H(P)| \le L^{2K} = t^{o(1)}$ and the marginal degree bounds were proved in Section 2. Since $W_c \le k^{1/16} S$ and $k = t^{o(1)}$, its contribution is $o(S)$. Average the bound for distinct pairs, using the two marginals μ, $W_b = o(S)$, and $S \to \infty$. This proves $T \le 2\mathbb{E}_\mu H^*_p + o(S) = o(S)$.

At a level in $\mathcal{C}$ declare an ordinary edge bad if its test is inconsistent or if $\nu_{H(p)}(R_{\mathbf{y}_q}) < 1 - k^{-2}$. Its integrated probability is $o(S)$, because Markov's inequality bounds the second contribution by $k^2 \int_{\mathcal{C}} \mathbb{E}_E[1 - \nu_{H(p)}(R_{\mathbf{y}_q})] = o(S)$. Fix a point p and this level. Write b_p for its conditional bad-edge probability, $\nu = \nu_{H(p)}$, $\zeta = k^{-2}$, and

$$\tau(\nu) = \|\nu_A\|_2^2 + \|\nu_{B_1}\|_2^2 - 1.$$

Let $\mathcal{Y}_p$ be the states of its good neighbors. Every such state has $\nu(R_y) \ge 1 - \zeta$, and the actual point state belongs to each R_y, by consistency. Let $d_P(p)$ be the score on $\mathcal{C}$ at this edge pair at a point, using its full bit probability ρ_p. We claim

$$d_P(p) + \tau(\nu) \ge \frac{\varpi}{4}\, 2\rho_p(1-\rho_p) - C(b_p + \zeta). \tag{6.14}$$

As before, a pair with a bad slot costs only $O(b_p)$; its probability is at most $2b_p$. The classification by intersections of the sets R_y proves the claim as follows.

If $\mathcal{Y}_p$ is empty, $b_p = 1$, while $d_P(p) \ge -1/2$ and $\tau(\nu) \ge -1$, so the claim holds for an absolute C.

If $\mathcal{Y}_p = \{(a, b)\}$, put $r = \nu_A(a)$, $s = \nu_{B_1}(b)$. Since $r + s \ge \nu(R_{(a,b)}) \ge 1 - \zeta$,

$$\tau(\nu) \ge r^2 + s^2 - 1 \ge \tfrac{1}{2}(1-\zeta)^2 - 1 \ge -\tfrac{1}{2} - \zeta.$$

Two good neighbors have the same state and bit, so $d_P(p) \geq 1 - \varpi - O(b_p)$. Their sum has the positive lower bound needed for (6.14).

If two states share a row, the intersection of their sets R_y is in that row, of ν-mass at least $1 - 2\zeta$. Thus $\tau(\nu) \geq (1 - 2\zeta)^2 - 1 \geq -4\zeta$. If every state is in that row, the actual point lies there too, since it belongs to the sets R_y for both distinct states. All good edges consequently have bit 1. Their full-law variation is at most $2b_p$, and $d_P(p) \geq -2b_p$, proving the claim in this case. Columns give bit 0 and the same estimates. If there is a third state outside the row or column, the three sets R_y have intersection of size at most one and mass at least $1 - 3\zeta$. It must be a singleton for large k. Both marginals then have an atom of at least that mass, giving

$$\tau(\nu) \geq 2(1 - 3\zeta)^2 - 1 \geq 1 - 12\zeta.$$

Together with $d_P(p) \geq -1/2$, this is more than sufficient.

Finally suppose no coordinate is shared by distinct states in $\mathcal{Y}_p$. Three such states are impossible, since the actual point belongs to all their sets R_y and their intersection is empty. In the remaining case there are two states differing in both coordinates. The two opposite corners carry ν-mass at least $1 - 2\zeta$. Each coordinate marginal therefore has at least that mass on a set of size two, so

$$\tau(\nu) \geq 2 \cdot \frac{(1 - 2\zeta)^2}{2} - 1 \geq -4\zeta.$$

The actual point is one of those corners. Its bit determines the good neighbor state; hence (6.10) gives $d_P(p) \geq \varpi/2 - O(b_p)$. This completes the proof of (6.14).

Average (6.14) over p and integrate on levels in $\mathcal{C}$. The bad-edge integral is $o(S)$, and $\zeta W_c \leq k^{-31/16} S = o(S)$. The result is

$$(\text{contribution on } \mathcal{C} \text{ to } \mathcal{D}_P) + T \geq \frac{\varpi}{4} C_P - o(S).$$

The contribution outside $\mathcal{C}$ to $\mathcal{D}_P$ is at least $-o(S)$ by Proposition 5.6, under the present assumptions on both exception means. Therefore

$$\frac{\varpi}{4} C_P - o(S) \leq \mathcal{D}_P + T \leq 10\delta^2 \varpi C_P + o(S),$$

and again $10\delta^2 < 1/4$ yields $C_P = o(S)$.

To finish, let V_P, V_Q be the variations for edge pairs at points and centers at any fixed level. Assign to each vertex its conditional majority bit, breaking ties in a fixed way. The probability that the bit on an incident edge differs from this label is at most the corresponding average variation, because $\min(\rho, 1 - \rho) \leq 2\rho(1 - \rho)$. Thus the two endpoint labels disagree on an edge with probability at most $V_P + V_Q$. When that sum is small, the matching-label consequence of expansion in Lemma 2.5 gives one bit label with total marginal exception probability $O(V_P + V_Q)$. The edge bit differs from it with probability $O(V_P + V_Q)$ as well. When the sum is not small the same bound follows from $f \leq 1/2$, with a larger absolute constant. Consequently

$$f \leq C(V_P + V_Q)$$

uniformly for all sufficiently large t. Integrating on levels in $\mathcal{C}$ gives $\int_{\mathcal{C}} f \leq C(C_P + C_Q) = o(S)$.

On levels in $\mathcal{G}$, the minority-bit probability is bounded by the two marginal exception probabilities and the inconsistency probability, as in Lemma 5.2. Its integral is $o(S)$ under the present assumptions. Levels in $\mathcal{B}$ contribute at most $W_b/2 = o(S)$. Thus $S = \int f = o(S)$, a contradiction. This proves (6.3). □

The positive exception mass on $\mathcal{G}$ is a first-moment statement. To extract many pairs at a single height scale, Section 7 next controls its tail of the exception mass and removes errors in measure weighted by exception mass before selecting a band. Throughout that selection, the bound $C_Q = o(S)$ for the bit-disagreement integral in experiment Q remains available under the original law of an edge pair at a center.

7 Extracting pairs with large ratio heights and small alternating-product heights

We now use Propositions 6.1 and 6.2: the bit-disagreement integral for experiment Q satisfies $C_Q = o(S)$, whereas the point exception mass satisfies $\liminf \mathbb{E}_\mu H_p^*/S > 0$. Here $S \to \infty$, and the parameters, levels, and exception indicators are those of Section 5. Our objective is a graph on two copies of P whose edges carry ratios of coordinate differences along unit edges. These ratios will have large height, while their alternating products around every 4-cycle have negligible height.

Throughout this section an *edge pair at a center* is sampled by taking q with law κ and then taking p, r independently and uniformly among its neighbors in E. Its two point marginals are μ. We retain the notation W_c for the width of levels in $\mathcal{C}$, W_b for the width of levels in $\mathcal{B}$, and m_{pq} for the integrated inconsistency indicator of the edge pq. In particular,

$$\mathbb{E}_\mu H_p^* + \mathbb{E}_\kappa H_q^* = O(S), \qquad \mathbb{E}_E m_{pq} = O(1), \qquad W_c \le S/\epsilon, \qquad W_b = o(S), \quad \epsilon = k^{-1/16}. \tag{7.1}$$

All integrals over levels include the prescribed place weights.

Proposition 7.1 (Pair graph with large ratio heights and small alternating-product heights)**.** *For the incidence graphs and levels constructed above, there are $J \to \infty$, a sequence $\omega_t \to 0$, and a simple bipartite graph $\mathcal{E}$ between two copies of P, each of size at most $t^{1+o(1)}$, with at least $t^{2-o(1)}$ edges, having the following properties. Each edge (p, r) joins distinct points and carries a ratio*

$$\lambda_{pr} = \frac{z_p - z_q}{z_r - z_q} \quad \text{for some } q \in Q \text{ with } pq, rq \in E.$$

No edge has both endpoints on any real unit circle containing at least $t^{9/10}$ points of P. Uniformly over the edges and 4-cycles of $\mathcal{E}$,

$$(z_p - z_r)(w_p - w_r) = 2 - \lambda_{pr} - \lambda_{pr}^{-1}, \tag{7.2}$$

$$h(\lambda_{pr}) \ge J/3, \tag{7.3}$$

$$h\left(\frac{\lambda_{pr}\lambda_{p'r'}}{\lambda_{pr'}\lambda_{p'r}}\right) \le \omega_t J. \tag{7.4}$$

In the last line all four indicated edges are required to be present; the witnessing centers may differ from edge to edge.

The proof is completed in Section 7.4. The tail bound Lemma 7.3 and function estimates Lemma 7.4 allow Lemma 7.5 to retain a common height scale after removing errors in measure weighted by exception mass. The deletion in Lemma 7.6 costs less than the retained probability under the law of an edge pair at a center. The atom bound (2.16) then converts that probability into a count of pairs, and the point functions cancel in each alternating product around a 4-cycle.

7.1 A pointwise estimate and its tail consequence

The first-moment bound in (7.1) alone does not prevent all exception mass from being carried by extremely rare points. We obtain the needed tail control by applying the product formula separately to each distinct pair of neighbors of a point.

Lemma 7.2 (Pointwise estimate outside $\mathcal{C}$)**.** *For every $p \in P$ and every pair of distinct neighbors $q, q' \in Q$,*

$$2H_p^* \le 3(H_q^* + H_{q'}^* + m_{pq} + m_{pq'}) + W_c + W_b + 2\log 2. \tag{7.5}$$

Proof. At a level in $\mathcal{G}$, let X, U, V be the exception indicators of p, q, q', and let I, I' be the inconsistency indicators of the two edges. Write P_+ for the positive charge on $\mathcal{G}$ in the score for an edge pair at a point: it equals two when p is exceptional, both bits are minority bits, and the two neighbors have matching nonshared coordinates; otherwise it is zero. Write $N = |Y_{pq} - Y_{pq'}|$. If $U = V = I = I' = 0$, the two bits are the same function of X: they are both majority bits for $X = 0$, and both minority bits for $X = 1$. In the latter case the nonshared coordinate of both neighbors has the dominant label, so $P_+ = 2$. Consequently, pointwise,

$$P_+ \ge 2X - 2(U + V + I + I'), \qquad N \le U + V + I + I',$$

and hence

$$P_+ - N \ge 2X - 3(U + V + I + I'). \tag{7.6}$$

In particular, a simultaneous exception at p and at a neighbor is charged directly to the neighbor indicator.

For the upper bound, apply the placewise argument in the proof of Proposition 5.4 to the distinct, nonzero coordinate differences $d = z_p - z_q$ and $d' = z_p - z_{q'}$. At a place v, write $b \le b'$ for their thresholds, let $\alpha = -\log|d - d'|_v - b$, and put $T_v = \alpha + c_v$, where $c_v = \log 2$ at infinity and zero otherwise. As shown there, $T_v \ge 0$, and all positive charges on $\mathcal{G}$ are confined to one of the intervals in (5.15), of length T_v. Their coefficients are at most two, so their total is at most $2T_v$, including when there are no positive charges on $\mathcal{G}$. The all-level negative charge is $b' - b$, and (5.14) therefore bounds the placewise score by

$$2T_v - (b' - b) = -\log|(d - d')(1/d - 1/d')|_v + 2c_v.$$

The product formula and total archimedean weight one now bound the positive charges on $\mathcal{G}$ minus the all-level negative charge by $2\log 2$.

Integrating (7.6) gives a lower bound for that same score. Outside the levels in $\mathcal{G}$ there are no positive terms, and the negative charge is at most $W_c + W_b$. Comparing the lower and upper bounds proves (7.5). □

Lemma 7.3 (Tail control)**.** *Let $R_0 = \epsilon^{-2} = k^{1/8}$. Then*

$$\mathbb{E}_\mu\big[H_p^* \mathbf{1}_{\{H_p^* > R_0 S\}}\big] = o(S). \tag{7.7}$$

Proof. Choose an absolute C_0 for which each of the two exception expectations in (7.1) is at most $C_0 S$, for all sufficiently large indices. Put

$$T = \{p : H_p^* > R_0 S\}, \qquad A_t = \mathbb{E}_\mu[H_p^* \mathbf{1}_T], \qquad d_{P,\min} = \min_{p \in P} d(p).$$

On T, the deterministic last three terms of (7.5) have sum at most H_p^*, eventually, since

$$\frac{W_c + W_b + 2\log 2}{R_0 S} \le \epsilon + \frac{W_b + 2\log 2}{R_0 S} = o(1).$$

After absorbing them, we may cap each neighbor exception mass at H_p^*:

$$H_p^* \le 3\big(\min(H_p^*, H_q^*) + \min(H_p^*, H_{q'}^*) + m_{pq} + m_{pq'}\big), \qquad p \in T, \tag{7.8}$$

for distinct q, q'. If a cap changes a term, that capped term already equals H_p^* and suffices for this inequality; otherwise the preceding inequality is unchanged.

Sample p with law μ and two neighbors independently and uniformly given p. Restrict the left side to distinct neighbors. The exact conditional probability of distinctness is $1 - 1/d(p)$. On the nonnegative right side we can drop this restriction. Thus

$$(1 - d_{P,\min}^{-1}) A_t \le 6\mathbb{E}_E[\min(H_p^*, H_q^*)\mathbf{1}_T] + C_m, \tag{7.9}$$

where C_m is an absolute upper bound for $6\mathbb{E}_E m_{pq}$. This calculation uses the distinctness probability at each individual point, so it remains valid even when its exception mass is very large.

We estimate the capped expectation by integrating threshold-exceedance indicators. For $u \ge 0$ let

$$F_u = \{p : H_p^* > \max(u, R_0 S)\}, \qquad G_u = \{q : H_q^* > u\},$$

and write $\phi_P(u) = \mu(F_u)$, $\phi_Q(u) = \kappa(G_u)$. The marginal version of the expansion estimate from Section 2, with $\eta = \gamma - 1/2 > 0$, is

$$\mathbb{P}_E(F, G) \le \sqrt{\mu(F)\kappa(G)}\left[\gamma^{-(1+2\eta)}(\mu(F)\kappa(G))^{\eta} + \gamma^{-1}\xi_t\right].$$

Because $\phi_P(u) \le C_0/R_0$ and $\phi_Q(u) \le 1$, the same multiplier

$$\vartheta_t = \gamma^{-(1+2\eta)}(C_0/R_0)^{\eta} + \gamma^{-1}\xi_t = o(1)$$

works at every threshold u, including thresholds where the center set has full measure. Interchange of nonnegative integrals and Cauchy–Schwarz now give

$$\begin{aligned}
\mathbb{E}_E[\min(H_p^*, H_q^*)\mathbf{1}_T] &= \int_0^\infty \mathbb{P}_E(F_u, G_u)\,\mathrm{d}u \\
&\le \vartheta_t \left(\int_0^\infty \phi_P(u)\,\mathrm{d}u\right)^{1/2} \left(\int_0^\infty \phi_Q(u)\,\mathrm{d}u\right)^{1/2} \\
&= \vartheta_t \sqrt{A_t \mathbb{E}_\kappa H_q^*} \le \vartheta_t \sqrt{C_0 A_t S}.
\end{aligned}$$

The first integral equals the unknown tail first moment A_t, including its contribution below the threshold $R_0 S$. Since $d_{P,\min} \to \infty$, (7.9) implies eventually

$$\tfrac{1}{2} A_t \le 6\vartheta_t \sqrt{C_0 A_t S} + C_m \le \tfrac{1}{4} A_t + 36 C_0 \vartheta_t^2 S + C_m.$$

The last step follows from nonnegativity of a square. Consequently

$$\frac{A_t}{S} \le 144 C_0 \vartheta_t^2 + \frac{4C_m}{S} \longrightarrow 0,$$

as required. □

7.2 Point functions and a common height band

We next express the ratio of coordinate differences for an edge pair at a center as a difference of functions attached to its two point endpoints. At a level $(v, s) \in \mathcal{G}$, let $I_x(v, s)$ be the exception indicator of the vertex x from Section 5, and let $\sigma(v, s) = 1$ when one is the minority bit and -1 when zero is the minority bit. Define a real-valued function of absolute-value entries by

$$a_p(v) = \int_{\{s:(v,s)\in\mathcal{G}\}} \sigma(v, s) I_p(v, s)\, \mathrm{d}s. \tag{7.10}$$

For a function of absolute-value entries b, write $\|b\|_1 = \sum_v^* |b(v)|$.

Lemma 7.4 (Functions associated with edge pairs at a center)**.** *On the space of edge pairs at a center there are nonnegative functions $E_1 = E_1(p, q, r)$ and $E_2 = E_2(p, q, r)$, with $\mathbb{E}E_1 + \mathbb{E}E_2 = o(S)$, such that every edge pair at a center satisfies*

$$\|(n_{pq} - n_{rq}) - (a_p - a_r)\|_1 \le E_1, \tag{7.11}$$

$$\|n_{pq} - n_{rq}\|_1 \ge H_p^* + H_r^* - E_2. \tag{7.12}$$

Here $n_{pq}(v) = -\log|z_p - z_q|_v$.

Proof. On $\mathcal{G}$, let I_{pq} be the levelwise inconsistency indicator. At a level in $\mathcal{G}$, write $b \in \{0, 1\}$ for the majority bit. Consistency gives

$$Y_{pq} = b + \sigma(I_p + I_q)$$

unless both endpoints are exceptional or the edge is inconsistent. In all cases the absolute residual in this identity is at most $2(I_p I_q + I_{pq})$. Subtracting the identities for the two edges cancels the center contribution. At each place,

$$n_{pq}(v) - n_{rq}(v) = \int (Y_{pq} - Y_{rq})\, \mathrm{d}s,$$

since both bits are single thresholds inside the level interval. It follows that (7.11) holds with

$$E_1 = \int_{\mathcal{C}} |Y_{pq} - Y_{rq}| + W_b + 2\int_{\mathcal{G}} (I_p I_q + I_r I_q + I_{pq} + I_{rq}).$$

Its mean is $o(S)$: the first term averages to $C_Q = o(S)$, the joint edge exceptions have integrated expectation $o(S)$, and the remaining bounds are in (7.1).

There is also the exact identity

$$\|n_{pq} - n_{rq}\|_1 = \int |Y_{pq} - Y_{rq}|.$$

On a level in $\mathcal{G}$ with exactly one exceptional point endpoint, a center with the dominant label, and two consistent edges, the bits differ. More explicitly, at every level in $\mathcal{G}$,

$$|Y_{pq} - Y_{rq}| \ge I_p + I_r - 2(I_p I_r + I_p I_q + I_r I_q + I_{pq} + I_{rq}).$$

If none of the terms subtracted is present, this is the preceding observation; if any is present, the right side is nonpositive. Integrating proves (7.12) with

$$E_2 = 2\int_{\mathcal{G}} (I_p I_r + I_p I_q + I_r I_q + I_{pq} + I_{rq}).$$

The integrated joint point exceptions on an edge pair at a center and joint edge exceptions are $o(S)$, by Lemma 5.2. Thus $\mathbb{E}E_2 = o(S)$ as well. □

The next selection makes these average error bounds uniform before choosing an interval with upper endpoint twice its lower endpoint. This order matters because the probability of the eventual band tends to zero.

Lemma 7.5 (A band of accurate edge pairs at a center)**.** *There are a constant $c > 0$, independent of t along the fixed sequence, a sequence $\delta_t \to 0$, and scales $J \to \infty$ such that a set of edge pairs at a center of probability at least*

$$\frac{c}{R_0 \log(2R_0)} = t^{-o(1)} \tag{7.13}$$

satisfies, uniformly on the set,

$$J \le H_p^* + H_r^* \le 2J, \qquad E_1 + E_2 \le \delta_t (H_p^* + H_r^*). \tag{7.14}$$

Every such edge pair at a center has $p \neq r$.

Proof. Set $W = H_p^* + H_r^*$ and $Z = E_1 + E_2$. By Proposition 6.2, choose $c_0 > 0$, independent of t along the fixed sequence, such that $\mathbb{E}W \ge 4c_0 S$ eventually. Choose $\delta_t \to 0$ sufficiently slowly that $\mathbb{E}Z/\delta_t = o(S)$. Before selecting any band, discard edge pairs at a center with $Z > \delta_t W$. Their lost weight is at most

$$\mathbb{E}[W \mathbf{1}_{\{Z > \delta_t W\}}] \le \frac{\mathbb{E}Z}{\delta_t} = o(S).$$

Also discard $W < c_0 S$, losing at most $c_0 S$. If $W > 2R_0 S$, its larger summand exceeds $R_0 S$ and is at least $W/2$. Thus Lemma 7.3 gives

$$\mathbb{E}[W \mathbf{1}_{\{W > 2R_0 S\}}] \le 2\mathbb{E}[H_p^* \mathbf{1}_{\{H_p^* > R_0 S\}}] + 2\mathbb{E}[H_r^* \mathbf{1}_{\{H_r^* > R_0 S\}}] = o(S).$$

The surviving weighted mass is at least $2c_0 S$, whereas every surviving edge pair at a center has $W \le 2R_0 S$. Its probability is therefore at least c_0/R_0.

Partition $[c_0 S, 2R_0 S]$ into at most $1 + \lceil \log_2(2R_0/c_0) \rceil = O(\log(2R_0))$ intervals with upper endpoint twice the lower endpoint, using half-open endpoints except at the final endpoint. One band has the probability in (7.13); if J is its lower endpoint, then $J \ge c_0 S \to \infty$, and (7.14) holds. The parameter hierarchy gives $k = t^{o(1)}$, so the lower bound in (7.13) is indeed $t^{-o(1)}$. Finally, if $p = r$, the left side of (7.12) is zero, whereas its right side is at least $(1 - \delta_t)W > 0$ for all sufficiently large indices. Such an edge pair at a center cannot survive. □

7.3 Removing common unit circles and choosing witnesses

The algebraic argument will require that a retained pair does not lie on a common unit circle containing many points of P. This exclusion must concern all real unit circles, including those whose centers do not belong to Q.

Consider real unit circles containing at least $t^{9/10}$ points of P.

Lemma 7.6 (Deleting pairs on circles containing at least $t^{9/10}$ points)**.** *The probability under the law of an edge pair at a center of p, r lying on a common unit circle containing at least $t^{9/10}$ points of P is at most*

$$t^{-3/5+o(1)} + t^{-9/5+o(1)} + \frac{2}{\min_{q \in Q} d(q)}. \tag{7.15}$$

This is $o(1/(R_0 \log(2R_0)))$. In particular, removing such edge pairs at a center from the set in Lemma 7.5 preserves the lower probability bound (7.13), after decreasing its constant.

Proof. Write M for the number of unit circles containing at least $t^{9/10}$ points of P. Every pair of distinct real points lies on at most two unit circles. Counting pairs on unit circles containing at least $t^{9/10}$ points of P therefore gives

$$M\binom{\lceil t^{9/10}\rceil}{2} \le 2\binom{|P|}{2}, \qquad M \le t^{1/5+o(1)},$$

using $|P| = t^{1+o(1)}$. The set P_{mult} of points on at least two unit circles containing at least $t^{9/10}$ points of P has size at most $2\binom{M}{2} \le t^{2/5+o(1)}$, because distinct circles intersect in at most two points. The degree bounds give

$$\max_p \mu(p) \le t^{-1+o(1)}, \qquad \max_q \kappa(q) \le t^{-2+o(1)}.$$

Consequently edge pairs at a center hitting P_{mult} at either point endpoint have probability at most $t^{-3/5+o(1)}$. Centers in Q whose own unit circle contains at least $t^{9/10}$ points of P number at most M; edge pairs at a center with such a center have probability at most $t^{-9/5+o(1)}$.

To bound the remaining bad edge pairs at a center, fix $p \notin P_{\mathrm{mult}}$ and a center q whose own circle contains fewer than $t^{9/10}$ points of P, in the original sampling law. The point p belongs to at most one unit circle containing at least $t^{9/10}$ points of P. If there is one, it differs from the unit circle centered at q, since the latter contains fewer than $t^{9/10}$ points of P. Their intersection has at most two points. In this original law, r is a uniform neighbor of q, so the probability that it shares a unit circle containing at least $t^{9/10}$ points of P with p is at most $2/d(q)$, even before requiring $r \notin P_{\mathrm{mult}}$. This proves (7.15).

For the comparison with the selected band, recall $R_0 = k^{1/8}$, $k = t^{o(1)}$, and the explicit requirement $k^{20}/\min_q d(q) \to 0$ in Lemma 5.1. The first two terms of (7.15), multiplied by $R_0 \log(2R_0)$, tend to zero by a fixed power of t. For the third term,

$$\frac{R_0 \log(2R_0)}{\min_q d(q)} = \frac{k^{1/8}\log(2k^{1/8})}{k^{20}} \frac{k^{20}}{\min_q d(q)} \longrightarrow 0.$$

This establishes the required relative loss bound. □

7.4 Choosing the pair graph

We can now forget the probability weights and retain an ordinary graph. The estimates already hold separately for every surviving edge pair at a center, so choosing one center for each endpoint pair preserves them.

Proof of Proposition 7.1. Take the band from Lemma 7.5, and remove the edge pairs at a center excluded by Lemma 7.6. Its probability remains at least $c/(R_0 \log(2R_0)) = t^{-o(1)}$, and every surviving edge pair at a center has distinct point endpoints. For any fixed off-diagonal pair (p, r), its probability under the law of an edge pair at a center is

$$\sum_{q:\, pq, rq \in E} \frac{1}{|E|\, d(q)} \le \frac{2}{|E| \min_q d(q)}.$$

Here the sum has at most two terms, since two distinct real points have at most two unit-circle centers. The retained edge pairs at a center therefore contain at least

$$\frac{c|E| \min_q d(q)}{2R_0 \log(2R_0)} \ge t^{2-o(1)}$$

distinct ordered endpoint pairs, using $|E| = t^{2+2a+o(a)}$ and $\min_q d(q) = t^{a+o(a)}$. These pairs form $\mathcal{E}$. Choose one surviving witnessing center for each pair.

For its two coordinate differences $d = z_p - z_q$, $d' = z_r - z_q$, the exact unit equations give $w_p - w_q = 1/d$ and $w_r - w_q = 1/d'$. Hence

$$(z_p - z_r)(w_p - w_r) = (d - d')(1/d - 1/d') = 2 - d/d' - d'/d,$$

proving (7.2). The product formula gives

$$2h(\lambda_{pr}) = \sum_v{}^{*} |\log|\lambda_{pr}|_v| = \|n_{pq} - n_{rq}\|_1 \geq (1 - \delta_t)J,$$

so (7.3) holds for all sufficiently large indices.

Finally, put

$$b_{pr}(v) = \log|\lambda_{pr}|_v + a_p(v) - a_r(v).$$

By (7.11) and (7.14), $\|b_{pr}\|_1 \leq 2\delta_t J$, uniformly on all retained edges. On a 4-cycle, the functions attached to the individual points cancel, leaving

$$\log \left| \frac{\lambda_{pr}\lambda_{p'r'}}{\lambda_{pr'}\lambda_{p'r}} \right|_v = b_{pr}(v) + b_{p'r'}(v) - b_{pr'}(v) - b_{p'r}(v).$$

Its weighted absolute sum is at most $8\delta_t J$. Another use of the product formula bounds its height by $4\delta_t J$, proving (7.4) with $\omega_t = 4\delta_t$. The exclusion of pairs on unit circles containing at least $t^{9/10}$ points of P holds by construction. □

8 The algebraic obstruction

The preceding section produced many pairs whose edge ratios have large height, although their alternating products around 4-cycles have negligible height. We now show that this is impossible. The argument uses a fixed complete bipartite pattern. Its positions will be sufficiently independent to separate several square roots, whereas differentiation of the relations for alternating products around 4-cycles will force a dependence between them.

Here and below a unit circle means a circle of radius exactly one in the original real plane. For a point $p = (x_p, y_p)$, retain the coordinates $z_p = x_p + iy_p$ and $w_p = x_p - iy_p$.

Proposition 8.1 (Obstruction to small alternating-product heights)**.** *There is no sequence of parameters $t_n, J_n \to \infty$, finite sets $P_n \subset \mathbb{R}^2$ of points with algebraic coordinates, and bipartite graphs $G_n \subset P_n \times P_n$, using two copies of P_n, with*

$$|P_n| \leq t_n^{1+o(1)}, \qquad |G_n| \geq t_n^{2-o(1)},$$

whose edges carry nonzero algebraic numbers λ_{pr} satisfying, for an absolute constant $c > 0$,

$$(z_p - z_r)(w_p - w_r) = 2 - \lambda_{pr} - \lambda_{pr}^{-1}, \tag{8.1}$$

$$h(\lambda_{pr}) \geq cJ_n, \tag{8.2}$$

$$h\left(\frac{\lambda_{pr}\lambda_{p'r'}}{\lambda_{pr'}\lambda_{p'r}}\right) = o(J_n) \tag{8.3}$$

uniformly over edges and 4-cycles, respectively, and such that no edge has both endpoints on a real unit circle containing at least $t_n^{9/10}$ points of P_n. Here h is the absolute logarithmic height defined before Lemma 4.2.

At this point every result used in the proof of Theorem 1.1 has been stated, and we give that deduction before proving Proposition 8.1.

Proof of Theorem 1.1. Suppose that no absolute power saving exists, and take the sequence of configurations obtained at the start of Section 2. Proposition 2.1 gives algebraic incidence subgraphs (P, Q, E) with the stated degree and expansion bounds. For these subgraphs, Proposition 4.3 shows that the weighted level integral defined in (4.8) tends to infinity. With the parameters of Lemma 5.1, Propositions 6.1 and 6.2 then give the vanishing bit disagreement on $\mathcal{C}$ and positive point-exception mass on $\mathcal{G}$ used in Proposition 7.1.

That proposition supplies a scale $J \to \infty$ and a graph $\mathcal{E}$ on two copies of P, with $|P| \le t^{1+o(1)}$ and at least $t^{2-o(1)}$ edges. Its algebraic edge ratios satisfy the pair identity (7.2), have absolute height at least $J/3$, and have alternating products around 4-cycles of height at most $\omega_t J$, where $\omega_t \to 0$, uniformly over all 4-cycles. No edge joins two points on a real unit circle containing at least $t^{9/10}$ points of P. Thus all the hypotheses of Proposition 8.1 hold, with $G_n = \mathcal{E}$, $t_n = t$, $J_n = J$ and $c = 1/3$. That proposition, proved in the rest of this section, gives a contradiction.

Hence the input sequence of Section 2 does not exist. By the argument at the start of that section, for some $j \ge 2$ every configuration of $n \ge 1$ points then has at most $j^2 n^{4/3-1/j}$ ordered unit pairs. Each unordered unit pair is counted twice, so Theorem 1.1 follows with $\beta = \max(1, 4/3 - 1/j) < 4/3$ and $C = j^2/2$. The cases $n = 0, 1$ are immediate. □

We prove Proposition 8.1 through a sequence of algebraic constructions. Assume its data exist, and suppress the index when this causes no ambiguity. At each index choose a finite normal number field F_n containing all coordinates, ratios and i. By Lemma 4.2, this choice does not affect the height. In the application from Proposition 7.1, take the field $\mathcal{N}$ already used there: every ratio is a quotient of coordinate differences.

8.1 The subfield defined by $h(u_n) = o(J_n)$

Fix a nonprincipal ultrafilter $\mathfrak{u}$ on $\mathbb{N}$: a collection of subsets closed under finite intersection and supersets, containing the cofinite sets and exactly one part of every finite partition. Such a collection is obtained by extending the cofinite filter to a maximal proper filter. Call a set of indices *large* if it belongs to $\mathfrak{u}$. Two sequences are identified when they agree on a large set. Write

$$F = \prod_{n \to \mathfrak{u}} F_n, \qquad \mathcal{U} = \prod_{n \to \mathfrak{u}} \mathbb{C},$$

where F_n is the number field just chosen. These are characteristic-zero fields, with $F \subset \mathcal{U}$: the inverse of a nonzero class is defined coordinatewise on its large set of nonzero entries. The field $\mathcal{U}$ is algebraically closed. Indeed, a polynomial of fixed positive degree has that degree on a large set, and choosing one complex root at each such index supplies a root in $\mathcal{U}$. Complex conjugation also acts coordinatewise on $\mathcal{U}$.

We use limits only along $\mathfrak{u}$. Thus $a_n = o(J_n)$ means that $\{n : |a_n| < \varepsilon J_n\}$ is large for every $\varepsilon > 0$. Edge counts of at least $t_n^{2-o(1)}$ are understood in this sense. Define

$$k_0 = \{[u_n] \in F : h(u_n) = o(J_n)\}.$$

Lemma 8.2. *The set k_0 is a subfield of F, and every element of F algebraic over k_0 belongs to k_0.*

Proof. The height inequalities

$$h(uv) \le h(u) + h(v), \qquad h(u+v) \le h(u) + h(v) + \log 2, \qquad h(u^{-1}) = h(u) \quad (u \ne 0)$$

prove the field assertion, since $J_n \to \infty$. These statements do not depend on representatives, and $h(0) = 0$. Constant rational sequences have bounded height, so $\mathbb{Q} \subset k_0$.

Suppose that $u \in F$ satisfies $u^m + a_{m-1}u^{m-1} + \cdots + a_0 = 0$ with $a_r \in k_0$ and fixed $m \geq 1$. At each finite index where the equation holds, the nonarchimedean triangle inequality gives $|u|_v \leq \max(1, \max_r |a_r|_v)$. At infinity the same estimate holds with the additional factor m: if $|u|_v > 1$, divide the equation by $|u|_v^{m-1}$. Since the total weight of the infinite places is one,

$$h(u) \leq \sum_{r=0}^{m-1} h(a_r) + \log m.$$

This estimate holds on a large set of indices. Its right side is $o(J_n)$, proving the claim. □

In particular, any edge ratio selected from the graphs is transcendental over k_0, by (8.2). The alternating products around 4-cycles in (8.3), in contrast, belong to $k_0^\times$. We next select finitely many edges whose positions retain enough algebraic independence to exploit this distinction.

8.2 A complete bipartite subgraph with algebraically independent coordinates

For every fixed pair of part sizes, Lemma 8.3 will give a complete bipartite pattern whose two coordinates at each plane point, and whose first coordinate at each curve point, are jointly algebraically independent over the field of locus coefficients. This independence will survive the later addition of the derivative data at at least one left point, by Lemma 8.5.

We first restrict the graph to plane or bounded-degree curve loci of minimum total dimension among those retaining at least $t_n^{2-o(1)}$ edges. This restriction will give uniform bounds for the vertices satisfying additional polynomial equations, allowing us to choose a complete bipartite subgraph whose limiting positions have no relations, beyond the locus equations, over the field generated by their coefficients.

We use two standard algebraic facts. First, for every field K, $K[Z, W]$ is a unique factorization domain [28, Lemma 10.120.10]. A polynomial in W with coefficients in $K[Z]$ is primitive if its coefficients have no common nonconstant factor. Gauss's lemma then identifies factorization of primitive polynomials in $K[Z][W]$ with factorization over $K(Z)$: clear denominators and cancel common prime factors of the coefficients. In particular, an irreducible polynomial in $K[Z, W]$ of positive W-degree remains irreducible in $K(Z)[W]$. Two primitive polynomials that are scalar multiples over $K(Z)$ differ by an element of $K^\times$.

Second, two distinct irreducible plane curves of degrees at most D, over an algebraically closed field, have at most D^2 common points. Indeed their projective closures have the same degrees and no common component, so Bézout's theorem bounds the sum of their intersection multiplicities by D^2 [11, Section 5.3]. Every affine intersection contributes at least one. We will apply this both over $\mathbb{C}$ at finite indices and over the algebraically closed field $\mathcal{U}$.

By a locus at an index we mean either the whole (z, w)-plane or the zero set of a nonconstant irreducible polynomial in $\mathbb{C}[z, w]$ whose degree has a fixed bound along the sequence. Its dimension is two or one, respectively. Among pairs of such locus sequences retaining at least $t_n^{2-o(1)}$ graph edges, choose one of minimal total dimension. A minimum exists since the pair of planes is available and the possible total dimensions are $2, 3, 4$. Finite partitions let us fix all degrees and normalize one nonzero coefficient of each curve equation on a large set. Denote the limiting loci in $\mathcal{U}^2$ by V_1, V_2, their dimensions by d_1, d_2, and the field generated over $\mathbb{Q}$ by their normalized equation coefficients and i by B_0. There are finitely many such coefficients, so B_0 is countable. It is a subfield of $\mathcal{U}$.

Each limiting curve equation is irreducible over $\mathcal{U}$. Otherwise a factorization into positive-degree factors would give, from finitely many coefficient sequences, a factorization at a large set of indices. The degrees of its factors are bounded by the degree of the equation, so there is no unbounded factorization to consider.

Each finite curve contains $t_n^{1-o(1)}$ retained real points: divide the lower bound $t_n^{2-o(1)}$ for the edge count by the upper bound for the other vertex set. Its image under $(z, w) \mapsto (\overline{w}, \overline{z})$ shares all these points. The bounded intersection fact forces the two curves to be equal on a large set. The limiting curve has the same symmetry. Neither coordinate projection is constant, since a curve with constant z would be a line $z = c$, whose symmetric image $w = \overline{c}$ is a different line. Thus every curve equation has positive degree in both variables.

The loci cannot both be the same unit circle. To check the exact meaning of this assertion, suppose their limiting equations, after normalization, were

$$(z - u)(w - v) = 1. \tag{8.4}$$

Equality of its finitely many coefficients holds at a large set of finite indices. The symmetry gives $v = \overline{u}$, again an exact coordinatewise identity on a large set. Hence the finite loci there are the same real unit circle. Both vertex sets on it have size $t_n^{1-o(1)} > t_n^{9/10}$, contradicting the deletion hypothesis in Proposition 8.1. This reasoning concerns equality in the ultraproduct; ordinary convergence of a radius to one would not imply (8.4).

Restrict the graph to these loci and prune it to minimum degree $t_n^{1-o(1)}$ on both sides. Explicitly, if its initial edge and total vertex counts are e_n, v_n, repeatedly remove vertices of degree less than $e_n/(2v_n)$. Fewer than $e_n/2$ edges are removed. The remaining graph still has at least $t_n^{2-o(1)}$ edges, both vertex sets have size $t_n^{1+o(1)}$, and the asserted minimum degree holds.

When points are adjoined to a field below, both of their coordinates are adjoined.

Lemma 8.3 (Uniform avoidance and algebraically independent coordinates)**.** *For every pair of fixed positive integers N, j, the pruned graphs contain a complete bipartite subgraph $K_{N,j}$ on a large set of indices so that its limiting vertices $p_1, \dots, p_N \in V_1$ and $r_1, \dots, r_j \in V_2$ have the following property. The family consisting of both coordinates of each vertex on a plane locus and the z-coordinate of each vertex on a curve locus has $Nd_1 + jd_2$ members, all algebraically independent over B_0. The other coordinates on a curve are algebraic over these coordinates and B_0. In particular,*

$$\operatorname{trdeg}_{B_0(r_1,\dots,r_j)} B_0(r_1, \dots, r_j, p_1, \dots, p_N) = Nd_1. \tag{8.5}$$

The selected coordinates and the ratios on all edges of this complete bipartite subgraph belong to F.

Proof. First, on a side whose locus is the plane, for each fixed degree bound D there is a $b_D > 0$ such that every nonzero polynomial of degree at most D, even with coefficients depending on n, vanishes at at most $t_n^{1-b_D}$ pruned vertices on a large set of indices. If this failed, choose at each index a polynomial maximizing its number of zeros among the finite vertex set. A maximum is attained because there are only finitely many subsets of that set. Failure of every fixed saving means that its zero count is $t_n^{1-o(1)}$ along $\mathfrak{u}$. One of at most D irreducible factors then contains $t_n^{1-o(1)}$ vertices. Their minimum degree supplies $t_n^{2-o(1)}$ edges after restricting this side to a bounded-degree curve, contradicting minimality. This proves the uniform assertion. On a curve side, a nonzero polynomial of degree D in the selected z-coordinate has at most D zeros among the vertices, since real points have distinct $z = x + iy$.

Let the two pruned set sizes be m, n' and their edge density be δ. Independent uniform vertices $p_1, \dots, p_N$ have a common-neighbor fraction X on the other side. Averaging degrees

there and applying Jensen gives $\mathbb{E}X \geq \delta^N$. Therefore the number of ordered complete bipartite configurations, initially allowing repetitions, is

$$m^N (n')^j \mathbb{E}X^j \geq m^N (n')^j \delta^{Nj} = t_n^{N+j-o(1)}. \tag{8.6}$$

Consider a fixed nonzero polynomial in the selected coordinates with coefficients in B_0. Choose representatives of its coefficients. Its vanishing among all ordered vertex tuples, before requiring edges, has a power saving:

$$\#\{\text{tuples at which the test vanishes}\} \leq t_n^{N+j-b} \tag{8.7}$$

on a large set, for some $b > 0$ depending on the test. Here is an induction that also handles moving coefficients. View the polynomial as a polynomial in the last point block, and select one nonzero coefficient polynomial in the earlier blocks. The earlier tuples annihilating this coefficient have a power saving by induction. At every other earlier tuple, the remaining polynomial is nonzero, with a fixed degree bound. The uniform plane estimate, or the bounded root count for a curve block, bounds its zeros. Multiply by the upper bounds $t_n^{1+o(1)}$ for the other block sizes and slightly decrease the saving. The starting case is exactly the one-block estimates. The finitely many coefficient identities and nonzero denominators involved hold on a large set.

There are countably many nonzero polynomial tests over B_0. Enumerate them as $T_1, T_2, \ldots$. Tuples with a repeated vertex within either part number at most $\binom{N}{2} m^{N-1} (n')^j + \binom{j}{2} m^N (n')^{j-1} = O_{N,j}(t_n^{N+j-1+o(1)})$, by the pruned vertex-set sizes. For each fixed s, this bound and the sum of (8.7) over the first s tests are negligible compared with (8.6) on a large set. Discarding these tuples therefore leaves a complete bipartite subgraph with distinct vertices within each part avoiding all the first s tests. At index n, choose the largest feasible prefix length $s_n \leq n$, and choose a complete bipartite subgraph avoiding that prefix. Include the validity of all representatives and denominators of a prefix in its feasibility requirement. If no prefix is feasible, use arbitrary number-field coordinates at that index. For every fixed s, the set where $n \geq s$ and the first s tests are feasible is large; hence $s_n \geq s$ on a large set. Each fixed polynomial test is therefore nonzero in the limiting complete bipartite subgraph. Only finite intersections of large sets were used.

For a curve equation $f(z, w) = 0$, its positive w-degree and the transcendence of z show that w is algebraic over $B_0(z)$. This proves all the transcendence assertions. The vertices and the assigned ratios at each chosen finite complete bipartite subgraph lie in F_n, so their classes lie in F. □

For a plane locus V over a field $\mathcal{B}$, define its function field to be $\mathcal{B}(V) = \mathcal{B}(z, w)$. For a curve with irreducible equation f, define it to be the fraction field of $\mathcal{B}[z, w]/(f)$. The coordinate condition used below for a point on V is transcendence degree $\dim V$ over $\mathcal{B}$.

We record explicitly how coordinates of transcendence degree equal to the locus dimension identify a function field. If $B_0 \subset \mathcal{B} \subset \mathcal{U}$ and a point on a locus V has transcendence degree $\dim V$ over $\mathcal{B}$, then evaluation at that point embeds $\mathcal{B}(V)$ into $\mathcal{U}$. For the plane this is algebraic independence. For a curve, its equation f is irreducible even over $\mathcal{U}$, hence over $\mathcal{B}$. Its coefficients when viewed as a polynomial in w have no common factor of positive degree in z. They cannot all vanish at an abscissa in $\mathcal{U}$: a common root would give a linear common factor over $\mathcal{U}$, contradicting irreducibility. If z were algebraic over $\mathcal{B}$, the nonzero specialized equation would make w algebraic as well. Thus z is transcendental. Gauss's lemma makes f irreducible over $\mathcal{B}(z)$, so every polynomial relation at the point is a multiple of f. Every nonzero denominator in the function field consequently evaluates to a nonzero element. The same argument applies to a point after adjoining the other vertices of the complete bipartite subgraph to $\mathcal{B}$, using Lemma 8.3.

8.3 Differentiation and a point retaining its transcendence degree

Take $N = 20$ and $j = 5$ in Lemma 8.3, and write $\lambda_{i\ell}$ for the edge ratios. All that is used of these values is $N > 3j$, in Lemma 8.5, and $j \geq 4$, in Section 8.5. Define

$$g_i = \lambda_{i1}, \qquad h_\ell = \frac{\lambda_{11}}{\lambda_{1\ell}}, \qquad \chi_{i\ell} = \frac{\lambda_{i\ell}\lambda_{11}}{\lambda_{i1}\lambda_{1\ell}}.$$

The condition on alternating products around 4-cycles gives $\chi_{i\ell} \in k_0^\times$, and

$$\lambda_{i\ell} = g_i h_\ell^{-1} \chi_{i\ell}. \tag{8.8}$$

Repeated rows or columns would violate the algebraic independence just proved; all vertices on either side are distinct.

A derivation is an additive map satisfying the product rule; being a k_0-derivation means that it vanishes on k_0.

Lemma 8.4. *There is a k_0-derivation $\partial : F \to F$ such that $\partial\lambda_{i\ell} \neq 0$ for every edge of the complete bipartite subgraph.*

Proof. Each $\lambda_{i\ell}$ is transcendental over k_0 by Lemma 8.2. Put it in a transcendence basis of F/k_0, prescribe its derivative to be one and those of the other basis elements to be zero, and extend by the quotient rule to the rational function field. Since the characteristic is zero, this derivation extends uniquely over the algebraic extension to F [7, Theorems 1.5 and 4.1]. For clarity, if α is algebraic over the rational function field, with minimal polynomial $P(X) = \sum_r c_r X^r$, this extension must satisfy

$$\partial\alpha = -\frac{\sum_r (\partial c_r)\alpha^r}{P'(\alpha)}.$$

The denominator is nonzero in characteristic zero; this is the extension formula in [7, Theorem B.8, equation (B.1)]. For an infinite algebraic extension, take the compatible extensions to its finite subextensions; uniqueness makes them agree on overlaps [7, Section 5].

We have thus obtained a witness derivation nonzero on each chosen ratio. The derivations form an F-vector space. A linear combination of the finitely many witnesses is nonzero on all ratios provided its coefficients avoid the zeros of a product of finitely many nonzero linear forms. This product is a nonzero polynomial over the infinite field F, so such coefficients exist. □

Put

$$\mathsf{a}_i = \frac{\partial g_i}{g_i}, \qquad \mathsf{a}'_\ell = \frac{\partial h_\ell}{h_\ell}, \qquad \mathbf{v}_i = (\partial x_{p_i}, \partial y_{p_i}), \qquad U_\ell = (\partial x_{r_\ell}, \partial y_{r_\ell}).$$

The derivation kills i, which is algebraic over $\mathbb{Q} \subset k_0$. Define

$$L_\ell(z, w) = (z - z_{r_\ell})(w - w_{r_\ell}), \qquad R_{i\ell} = \lambda_{i\ell}^{-1} - \lambda_{i\ell}.$$

Since $\partial\chi_{i\ell} = 0$, differentiating (8.8) gives

$$\frac{\partial\lambda_{i\ell}}{\lambda_{i\ell}} = \mathsf{a}_i - \mathsf{a}'_\ell, \qquad \partial(2 - \lambda_{i\ell} - \lambda_{i\ell}^{-1}) = (\lambda_{i\ell}^{-1} - \lambda_{i\ell})(\mathsf{a}_i - \mathsf{a}'_\ell).$$

Also $(z_p - z_r)(w_p - w_r) = (x_p - x_r)^2 + (y_p - y_r)^2$; the product rule therefore gives

$$2(p_i - r_\ell) \cdot (\mathbf{v}_i - U_\ell) = R_{i\ell}(\mathsf{a}_i - \mathsf{a}'_\ell), \tag{8.9}$$

$$R_{i\ell}^2 = L_\ell(p_i)(L_\ell(p_i) - 4), \tag{8.10}$$

$$\mathsf{a}_i - \mathsf{a}'_\ell \neq 0. \tag{8.11}$$

Here the dot denotes the bilinear expression $xu + yv$ in the ordinary two coordinates, with no conjugation. The second identity follows by squaring $\lambda^{-1} - \lambda$, and the third by Lemma 8.4.

We now form a coefficient field containing the right-hand positions and their derivative data. Let K_0 be the subfield of $\mathcal{U}$ consisting of all elements algebraic over

$$B_0(r_1, \dots, r_j, U_1, \dots, U_j, \mathsf{a}'_1, \dots, \mathsf{a}'_j).$$

This field is algebraically closed: roots in $\mathcal{U}$ of polynomials over it are still algebraic over the displayed field. We do not adjoin k_0 to K_0.

Lemma 8.5. *For some $i \in \{1, \dots, N\}$, the point $p = p_i$ has transcendence degree d_1 over K_0, and evaluation at this point gives a K_0-isomorphism $K_0(V_1) \simeq K_0(p)$.*

Proof. Over $B_0(r_1, \dots, r_j)$, all the left positions jointly have transcendence degree Nd_1, by (8.5). The right coordinate derivatives and scalars consist of at most $3j$ elements. Adjoining them and then taking algebraic closure can reduce that transcendence degree by at most $3j$. To see the bound, adjoining these elements and all left positions together gives an extension of $B_0(r_1, \dots, r_j)$ of transcendence degree at least Nd_1. In the tower through the field generated by the right data, the first step has transcendence degree at most $3j$. Subtracting it leaves at least $Nd_1 - 3j$. Passing from that intermediate field to its algebraic closure inside $\mathcal{U}$ does not change this remaining transcendence degree. Therefore

$$\operatorname{trdeg}_{K_0} K_0(p_1, \dots, p_N) \geq Nd_1 - 3j. \tag{8.12}$$

If no individual point had degree d_1, subadditivity would bound the left side by $N(d_1 - 1)$. This contradicts $N > 3j$. The function-field assertion is the evaluation argument following Lemma 8.3, with $\mathcal{B} = K_0$. □

The coordinate-derivative data may depend arbitrarily on the left positions. The degree count above allows those correlations.

We have reduced the proof to one point p of coordinate transcendence degree d_1 over K_0. Write $M = K_0(V_1) = K_0(p)$, $R_\ell = R_{i\ell}$, and

$$D_\ell = L_\ell(L_\ell - 4) \in M.$$

The next step proves directly over this enlarged field that these radicands are independent modulo squares.

8.4 Valuations taking value one on one radicand and zero on the others

An integer-valued valuation of M is a map $v : M^\times \to \mathbb{Z}$ with $v(ab) = v(a) + v(b)$ and $v(a + b) \geq \min\{v(a), v(b)\}$, where $v(0) = +\infty$. Our valuations will count the order of a zero at a chosen point or along an irreducible polynomial. The aim is to give each radicand one simple zero that none of the others shares.

Lemma 8.6. *For every $\ell \in \{1, \dots, j\}$ there is an integer-valued valuation v_ℓ of M such that*

$$v_\ell(D_\ell) = 1, \qquad v_\ell(D_m) = 0 \quad (m \neq \ell).$$

Consequently the classes of $D_1, \dots, D_j$ are independent in $M^\times/(M^\times)^2$.

The valuations are constructed in Section 8.6, over the same field $M = K_0(V_1)$. We first use their conclusion to finish the obstruction.

8.5 An unused sign change

We use the quadratic case of Kummer theory over the characteristic-zero field M. If $D_1, \dots, D_j \in M^\times$ have independent classes modulo squares and $R_\ell^2 = D_\ell$, then

$$[M(R_1, \dots, R_j) : M] = 2^j,$$

and every choice of signs of the R_ℓ is induced by an automorphism fixing M [20, Theorem 5.30 and Remark 5.32]. Here the subgroup in the theorem is $(M^\times)^2\langle D_1, \dots, D_j\rangle$, whose quotient by $(M^\times)^2$ has order 2^j. The theorem gives the degree, and its perfect pairing sends an automorphism σ to the character $[D_\ell] \mapsto \sigma(R_\ell)/R_\ell$. Since the square classes form a basis of this quotient, these characters realize every assignment of signs. The hypotheses hold because M contains both square roots of unity and has characteristic different from two. Independence of the square classes is required in this same field M, as proved in Lemma 8.6.

Apply this fact with the radicands of Lemma 8.6, and let $\mathcal{M} = M(R_1, \dots, R_j) \subset \mathcal{U}$, using the actual roots from (8.10). Write $\mathbf{v} = \mathbf{v}_i$, $\mathsf{a} = \mathsf{a}_i$ for the surviving point. We have the equations

$$2(p - r_\ell) \cdot \mathbf{v} - R_\ell \mathsf{a} = 2(p - r_\ell) \cdot U_\ell - R_\ell \mathsf{a}'_\ell, \qquad 1 \le \ell \le j. \tag{8.13}$$

Their right-hand positions, coordinate derivatives and scalars belong to K_0.

First suppose the three rows $p - r_1, p - r_2, p - r_3$ have rank two. The determinant of the first three equations in the unknowns v_x, v_y, a is a linear combination of R_1, R_2, R_3 over M. More explicitly, with $\Delta_{\ell m} = \det(p - r_\ell, p - r_m)$, it is

$$-4\big(R_1\Delta_{23} - R_2\Delta_{13} + R_3\Delta_{12}\big).$$

Rank two makes at least one minor nonzero. If $\sum_{\ell=1}^{3} a_\ell R_\ell = 0$ with $a_\ell \in M$, changing only the sign of R_m and subtracting gives $2a_m R_m = 0$. Hence all a_m vanish, proving the needed linear independence and the nonzero determinant. Cramer's rule places the existing solution v_x, v_y, a in $M(R_1, R_2, R_3)$. The automorphism of $\mathcal{M}$ that changes only the sign of R_4 fixes these unknowns. Apply it to the fourth equation in (8.13) and subtract. Since $R_4 \neq 0$, it follows that $\mathsf{a} = \mathsf{a}'_4$, contradicting (8.11).

We verify when the rank condition can fail. If V_2 is a plane or a curve other than a line, the three right-hand points with jointly algebraically independent selected coordinates are noncollinear. Indeed the first two are distinct, and a third point of coordinate transcendence degree d_2 over the field generated by B_0 and the first two points can lie on their line only if its locus is contained in that line, by the function-field evaluation embedding. Three noncollinear right-hand points make the three position rows have rank two for every p. If V_2 is a line, its first two right-hand points are distinct. The rows have rank two unless p belongs to that line. Since p has coordinate transcendence degree d_1 over K_0, this last possibility requires $V_1 = V_2$ to be the same line.

In that remaining case choose a nonzero direction vector $v \in K_0^2$ for the line, and write $p - r_\ell = s_\ell v$, with $s_\ell \in M^\times$. Nonzero row factors follow from distinctness of the points. Only the scalar $h = v \cdot \mathbf{v}$ enters (8.13). The first two equations in h, a have determinant

$$2(s_2 R_1 - s_1 R_2) \neq 0,$$

again by independence of the single radicals. Thus $h, \mathsf{a} \in M(R_1, R_2)$. Changing only the sign of R_3 in the third equation gives $\mathsf{a} = \mathsf{a}'_3$, the same contradiction.

This proves Proposition 8.1. All sign changes were applied inside the finite radical field $\mathcal{M}$, after Cramer's rule located the unknowns there.

8.6 Construction of the valuations

Proof of Lemma 8.6. An odd valuation value detects a nonsquare because the valuation of a square is even. We construct the stated valuations over K_0 in each possible locus case. All noncoincidences used in the constructions will first be proved from the original algebraic independence of the selected coordinates over B_0.

The left locus is the plane. The polynomial

$$Q_\ell = (z - z_{r_\ell})(w - w_{r_\ell}) - 4$$

is irreducible in $K_0[z, w]$. It is primitive as a polynomial in w, because its leading coefficient $z - z_{r_\ell}$ is coprime to its constant coefficient, and is linear over $K_0(z)$. The polynomials Q_ℓ are distinct, as comparison of the coefficients of z and w recovers their centers. None divides any $L_m = (z - z_{r_m})(w - w_{r_m})$. The exponent of Q_ℓ in a rational function is therefore the required valuation.

The left locus is a curve, and either the loci differ or its equation has degree at least two in w. Let $f(z, w)$ be its irreducible equation and set $a = z_{r_\ell}$. The coordinate a is transcendental over B_0. Consequently the leading coefficient of f in w does not vanish at a, and $f(a, w)$ has simple roots: the coprimality of f, f_w over $B_0(z)$, after clearing denominators, excludes multiple roots at this transcendental abscissa. In detail, irreducibility and characteristic zero give

$$\Phi(z, w)f(z, w) + \Psi(z, w)f_w(z, w) = q_*(z) \neq 0$$

for some $\Phi, \Psi \in B_0[z, w]$ and $q_* \in B_0[z]$. Since $q_*(a) \neq 0$, a root of $f(a, w)$ cannot be a root of $f_w(a, w)$. The nonvanishing leading coefficient preserves the degree in w, so all its roots are accounted for.

If $V_2 \neq V_1$, the right-hand point r_ℓ is not on V_1. Indeed its coordinate transcendence degree over B_0 equals $\dim V_2$, where V_2 is either the plane or a different irreducible curve. Choose any root b of $f(a, w)$. Then $b \neq w_{r_\ell}$. If $V_2 = V_1$ and f has degree at least two in w, then $f(a, w)$ has at least two distinct roots; choose one other than w_{r_ℓ}, again obtaining $b \neq w_{r_\ell}$. In both cases

$$s_\ell = (a, b) \in V_1, \qquad f_w(a, b) \neq 0, \qquad b \neq w_{r_\ell}.$$

The ordinate b is algebraic over $B_0(a)$. Since K_0 contains a and is algebraically closed, $b \in K_0$.

For $m \neq \ell$, the original algebraic independence of the selected coordinates implies that r_m still has coordinate transcendence degree d_2 over $B_0(s_\ell)$. Adjoining the ordinate b is an algebraic extension of adjoining a, so it does not reduce the transcendence degree of this independent right-hand point. Neither coordinate of r_m equals the corresponding coordinate of s_ℓ: both projections of a curve are nonconstant, and the assertion for a plane is immediate. Thus $L_m(s_\ell) \neq 0$. Also $L_m(s_\ell) \neq 4$. Otherwise the function-field evaluation embedding would force V_2 to lie in

$$(z - a)(w - b) = 4.$$

The plane cannot do so. If V_2 is a curve, this irreducible circle equation must be proportional to its defining equation. Normalize the coefficient of zw to one. The coefficient of w then recovers $-a$ from coefficients in B_0, contradicting the transcendence of a over B_0. These are nonvanishing statements in $\mathcal{U}$; placing more of their entries in K_0 cannot change them.

To obtain the valuation over K_0, introduce an indeterminate T and solve

$$f(a + T, b + c_1T + c_2T^2 + \cdots) = 0$$

recursively in $K_0[[T]]$. At each step the coefficient of the new unknown is $f_w(a,b) \neq 0$, so the solution exists uniquely. Indeed, after $c_1, \dots, c_{m-1}$ have been fixed, let b_m be the coefficient of T^m in $f(a+T, b+\sum_{r<m} c_r T^r)$. Adding $c_m T^m$ to the second argument changes that coefficient by $f_w(a,b)c_m$, so set $c_m = -b_m/f_w(a,b)$. This gives an embedding $K_0(V_1) \hookrightarrow K_0((T))$. Indeed z maps to the transcendental element $a+T$, and irreducibility of f over $K_0(z)$ shows that its only polynomial kernel is (f). Now

$$L_\ell = T\big(b - w_{r_\ell} + O(T)\big)$$

has order one, while $L_\ell - 4$ and all $L_m(L_m - 4)$, $m \neq \ell$, have nonzero constant terms. The order in T is the required valuation.

Both loci are the same curve, whose equation has degree one in w. The symmetry of this curve interchanges the two variable degrees of its irreducible equation. Thus its degree in z is one as well, and the equation has degree one in each variable. Solving for w gives either

$$w = ez + e', \qquad e \neq 0, \tag{8.14}$$

or

$$(z-u)(w-v) = b, \qquad b \neq 0. \tag{8.15}$$

The constants here belong to B_0. To see the classification directly, write the equation as $azw + cz + dw + e_0 = 0$. If $a = 0$, both linear coefficients are nonzero. If $a \neq 0$, complete the product; its constant is nonzero by irreducibility. In (8.15), $b \neq 1$, by the exact unit-circle exclusion (8.4).

In the line case, writing $a_\ell = z_{r_\ell}$,

$$L_\ell = e(z - a_\ell)^2.$$

The line has nonconstant projections, and its direction in the ordinary two coordinates has nonzero squared length proportional to $e \neq 0$. Thus $L_\ell - 4$ has two distinct simple zeros. Choose one, say $z = a_\ell + c$ with $c^2 = 4/e$, and let s_ℓ be the corresponding point on the line. Its coordinates are algebraic over $B_0(a_\ell)$. For an independent right-hand point r_m, the conditions $L_m(s_\ell) = 0$ or 4 would make a_m one of finitely many elements algebraic over this field. This contradicts the joint algebraic independence of the selected coordinates. Since $L_\ell = 4$ at the chosen zero, order at that abscissa in the rational function field $K_0(z)$ gives the required valuation.

In the fractional-linear case, put $\zeta = (z-u)/(a_\ell - u)$, where $a_\ell - u \neq 0$. Direct substitution gives

$$L_\ell = -b\frac{(\zeta - 1)^2}{\zeta}. \tag{8.16}$$

The equation $L_\ell - 4 = 0$ is equivalent to

$$b\zeta^2 + (4 - 2b)\zeta + b = 0.$$

Its discriminant is $16(1-b) \neq 0$, and its two roots are nonzero because $b \neq 0$. Hence they give simple zeros of $L_\ell - 4$ at finite points with $z \neq u$. The corresponding point s_ℓ is algebraic over $B_0(a_\ell)$. As a function of the other right-hand point abscissa a_m, its squared distance to that right-hand point is

$$-b\frac{(z_{s_\ell} - a_m)^2}{(z_{s_\ell} - u)(a_m - u)}.$$

This is a nonconstant rational function: its numerator has degree two and its denominator degree one, and $z_{s_\ell} \neq u$. Its values cannot be zero or four at the transcendental a_m over $B_0(s_\ell)$. Thus

the simple zero is shared by no other radicand, and order at its abscissa again gives the valuation over $K_0(z)$.

These cases exhaust the possibilities. In each, a nonempty product $\prod_{\ell\in I} D_\ell$ has valuation one at any v_ℓ with $\ell \in I$, and therefore is not a square. □

The excluded value $b = 1$ illustrates why the deletion of unit circles containing at least $t_n^{9/10}$ points of P_n was necessary. In that case

$$L_\ell(L_\ell - 4) = \left(\frac{\zeta^2 - 1}{\zeta}\right)^2.$$

Thus the valuations taking value one on one radicand and zero on the others would fail on a common unit circle. For all the retained cases they were constructed after passing to K_0.

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